UnionPay Integrated Circuit Card Specifications

—— Auxiliary Specifications

Part III Debit/Credit Application - Member Implementation Guide for Acquirer

Version 2017

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Table of Contents

Summary of Revisions

The change listed below is associated with the current version.

Change Description	Revised/ Deleted/ Added	Location
Added a description to specify the merchant information accuracy requirement.	Added	12.2.2

1 Application Scope

This book applies to all UPI Participants.

2 References

The terms and conditions of the following documents quoted in this guide have become the terms and conditions of the guide. The modification list (excluding corrected contents) or revised edition attached to the dated documents shall not apply to this Specification. However, Participants may study whether to apply the latest versions of such documents. The latest versions of non-dated documents shall apply to the guide.

EMV - Books 1 to 4 of the EMV 4.1 Integrated Circuit Card Specifications for Payment Systems

3 Terms and Definitions

The following terms and definitions are applied in this guide.

qUICS

Quick debit/credit application (qUICS) ensures quick transaction through contactless interface.

Financial IC card Debit/Credit Applications Root CA

Root certificate authority serves financial IC card security, hereinafter referred to as "Root CA". The system to implement the root CA functions is the "Financial IC card Debit/Credit Applications Root CA System", hereinafter referred to as "Root CA System".

Application

Application protocols and relevant data sets which are used between cards and terminals.

Clearing

A process where the issuer processes the transaction data submitted by the acquirer. As to the issuer, clearing provides another chance to synchronize the back-office counter with the counter embedded in cards. As to the acquirer, clearing is to submit the transaction data to the issuer.

Command

A message sent from the terminal to IC cards to initiate one operation or request a response.

Cryptogram

Result of cryptographic operation.

Electronic Cash (EC)

Function supporting low-value payment based on debit/ credit application.

Electronic Cash Balance

A counter that shows the available offline spending amount on the card.

Electronic Cash Balance Limit

The maximum amount cardholder can spend offline.

Financial Transaction

The commodity or service exchanged among cardholder, merchant and acquirer based on the ways of collection and payment.

Function

A sequence combined by one or more commands, which complete the whole or part of transaction.

Integrated Circuit (IC)

Electronic components designed to perform processing and/or memory functions.

Integrated Circuit(s) Card (IC card)

A card into which one or more integrated circuits are inserted to perform processing and memory functions.

Interface Device

That part of the terminal into which the IC card is inserted, including mechanical and electrical parts.

Issuer Action Code

The actions selected by issuer to be taken based on the analysis of Terminal Verification Result.

Key

A sequence of symbols that controls the operation of a cryptographic transformation.

Load

A process to increase electronic cash balance in the card. There are a variety of ways to load, such as transfer from main account, cash deposit or transfer from other accounts. Electronic cash balance cannot exceed electronic cash balance limit.

Magnetic-stripe

The stripe containing magnetically encoded information.

Path

Concatenation of file identifiers without delimitation.

Payment System Environment

The set of logical conditions established within the ICC when a payment system application conforming to UICS has been selected, or when a Directory Definition File (DDF) used for payment system application purposes has been selected.

Private Key

That key of an entity's asymmetric key pair that should only be used by that entity. In the case of a digital signature scheme, the private key defines the signature function.

Public Key

That key of an entity's asymmetric key pair that can be made public. In the case of a digital signature scheme, the public key defines the verification function.

Public Key Certificate

The public key information of an entity signed by the certification authority and thereby rendered unforgeable.

Response

A message returned by the ICC to the terminal after the processing of a command message received by the ICC.

Script

A Command or command sequence transmitted to terminal by issuer for the purpose of being sent serially to the ICC as commands.

Terminal

The device used in conjunction with the ICC at the point of transaction to perform financial transactions. It includes interface device, and also includes some other components and interfaces such as host communications.

Terminal Action Code

Terminal action code (default, decline and online) reflects the action selected by acquirer based on Terminal Verification Result.

Unload

Process of reducing electronic cash balance to zero. The result of unload is to transfer all remainder amount in electronic cash account to cardholder's main account or return the equivalent amount in cash. Unload is used when cardholder or Issuer cancels electronic cash function.

4 Symbols and Abbreviations

The following abbreviations and symbols are only applicable to this specification:

AAC	Application Authentication Cryptogram
AAR	Application Authorization Referral
AC	Application Cryptogram
ADA	Application Default Action
ADF	Application Definition File
AEF	Application Elementary File
AFL	Application File Locator
AID	Application Identifier
AIP	Application Interchange Profile
APDU	Application Protocol Data Unit
ARPC	Authorization Response Cryptogram
ARQC	Authorization Request Cryptogram
ATC	Application Transaction Counter
ATM	Automated Teller Machine
ATS	Answer To Select
AUC	Application Usage Control
BER	Basic Encoding Rules
BIN	Bank Identification Number
CA	Certificate Authority
CAM	Card Authentication Method
CDA	Combined DDA/AC Generation
CDOL	Card Risk Management Data Object List
CID	Cryptogram Information Data
CLA	Class Byte of the Command Message
cn	compress numeric
C-TPDU	Command TPDU
CTTA	Cumulative Total Transaction Amount
CTTAL	Cumulative Total Transaction Amount Limit

CTTAUL	Cumulative Total Transaction Amount Upper Limit
CVM	Cardholder Verification Method
CVR	Card Verification Results
CVN	Card Verification Number
dCVN	Dynamic Card Verification Number, a dynamic signature generated by card applications. Dynamic CVN replaces the
DDA	Dynamic Data Authentication
DDF	Directory Definition File
DDOL	Dynamic Data Authentication Data Object List
DES	Data Encryption Standard
DF	Dedicated File
DIR	Directory
DOL	Data Object List
GPO	GET PROCESSING OPTIONS
EC	Electronic Cash
EF	Elementary File
EMV	Europay MasterCard VISA
FCI	File Control Information
fDDA	Fast DDA
FWI	Frame Waiting time Integer
IAC	Issuer Action Code
IC	Integrated Circuit
IC Card	Integrated Circuit Card
iCVN	Integrated Circuit Card Verification Number
IDD	Issuer Discretionary Data
Lr	Length of Response Data Field
M	Mandatory
MAC	Message Authentication Code
MDK	Master DEA Key
MF	Mater File

n	Numeric
O	Optional
P1	Parameter 1
P2	Parameter 2
P3	Parameter 3
PAN	Primary Account Number
PDOL	Processing Options Data Object List
PKI	Public Key Infrastructure
PIN	Personal Identification Number
PIX	Proprietary Application Identifier Extension
PPSE	Proximity Payment Systems Environment, a table of supported application ids, application tags, and application priority indicators that can be accessed through contactless interface. This table includes entries to all directories, to be returned by the card in FCI response SELECT PPSE
PSE	Payment System Environment
qUICS	quick UICS to ensure fast transactions through contactless
RFU	Reserved for Future Use
RID	Registered Application Provider Identifier
RSA	A type of asymmetric key method created by Rivest, Sharmir, and Adleman for encryption and authentication
R-TPDU	Response TPDU
SAD	Signed Static Application Data
SDA	Static Data Authentication
SFI	Short File Identifier
SW1	Status Word One
SW2	Status Word Two
TAC	Terminal Action Code
TC	Transaction voucher
TDOL	Transaction voucher Data Object List
TLV	Tag Length Value

TSI	Transaction Status Information
TVR	Terminal Verification Results
UDK	Unique DEA Key
Proprietary	Not defined in this specification or/and beyond the scope of this specification
Must	Indicates mandatory requirement
Shall	Indicates recommended requirement

5 Terminal Selection and Authentication

5.1 Terminal Selection Criteria

5.1.1 Terminal Selection Overview

Prior to discussion with vendors and evaluating their products, acquirer shall start the process of compiling terminal requirements, and determine the minimum/mandatory/recommended terminal requirements according to acceptance environment.

This section summarizes the information related to acquirer terminal selection criteria for acquirers, to ensure products meet them:

- Specification requirements that Acquirer must ensure to meet for domestic and international standards compatibility:
  - ✓ Can read magstripe.
  - ✓ Can read chip. If the card has IC chips, terminals with chip readers are not allowed to use magstripe while skipping chip-authorized control intentionally; only if the chip is damaged or the chip/chip reader is not operational, could magstripe be used for transaction.
  - ✓ Support proprietary requirement.
  - ✓ Support transaction type requirement, e.g.: pre-authorization, cancellation, refund etc.
  - ✓ Support magstripe terminal requirement, e.g.: fallback processing, compatible requirement of identifying service code.
  - ✓ Support other application requirement, e.g.: loyalty program, stored value schemes.
  - ✓ Support cardholder account selection.

Confirm supported terminal types, attended/unattended, printing capable/ no print capability, handheld/desktop/wall-installation, internal PINPAD/external PINPAD/no PINPAD, contactless IC read-writer/no contactless IC read-writer. Common terminals include: modular terminal, integrated POS, ATM, handheld POS terminal, unattended terminal, self-service terminal, telephone payment terminal, and IC card offline acquirer terminal, etc.

Confirm whether terminal supports: standard credit and debit, electronic cash, contactless low-value payment, electronic cash application extension, dual currency electronic cash.

Identify that if terminal is online only or offline only, or online/offline capable.

Confirm whether the terminal needs to have digital signature capability.

Determine terminal requirement on exception files.

Confirm terminal requirements for backing up of offline transaction data.

Determine requirement of application selection (directory selection and AID list selection method).

Determine requirement of cardholder verification according to the target market (online PIN, offline plaintext/cryptogram PIN, signature etc).

Evaluating the product price based on the function, hardware, software, warranty clause, service support protocol of the device.

Requirements for terminal software: must provide high-level language (C language) development environment, provide dedicated interface for secondary development, provide application module, and has environment for application debugging and testing. Verify product functionality. Determine the requirement on transaction speed, and evaluate the supported transaction volume.

Determine which portion of merchant configurations are affected by chip and identify:

- ✓ Which can be realized through software modification
- ✓ Which require adding hardware;
- ✓ All the related vendors.

Judge if the current device can be updated, or require new supplementary device.

Determine any additional technical function requirement, e.g. radio frequency support, mobile-phone function, or internet connection.

Determine chip's impact on the floor limit, different floor limits that can be set for chips and magstripe.

5.1.2 Basic Terminal Requirements

Terminals must meet the following specifications:

- EMV Level 1 and Level 2 Specifications
- UnionPay Integrated Circuit Card Specification
- UnionPay Integrated Circuit Card Specification -- Basic Specification
- UPI Operating Regulations

Note: Relevant specifications will be periodically updated, please contact UnionPay representative to obtain the latest version;

5.1.3 Terminal Certification

Acquirer must choose terminal products that passed UPI card acquirer terminal product network certification.

Terminal products that passed UPI card acquirer terminal product network certification fall into different categories, including functionality support for magstripe cards, contact IC cards, or contactless IC cards. For terminals that support contact IC card functionality, the pre-requisite to passing certification is passing UICS/EMV Level 1 and Level 2 test certification; for terminals that support contactless IC card functionality, the pre-requisite is passing qUICS test certification.

5.1.4 IC Card Function Modularization

Terminal software structure should be configured so that IC card applications are isolated from other applications, with this modular structure recommended because: it is favorable to commercial investment protection and makes software upgrades easier. Maintaining the independence between IC card functionality and non-IC card functionality makes it so that when payment application needs to be upgraded or modified, IC card functionality will not be affected.

When developing new products on the basis of terminal IC card Level 1 and Level 2 functionality, there is no need for UICS/EMV Level 1 and Level 2 re-testing.

Under any circumstance, if terminal IC card functionality is affected, supplier must re-submit UICS/EMV certification, otherwise inter-operability problems is likely to result. Functional changes should be realized through parameterized configurations as must as possible, so that for a completely modular structural configuration terminal, a change to TAC will not impact its UICS /EMV compatibility; otherwise, Level 1 and 2 recertification may be required.

The vast majority of IC card functionality are mandatory, and terminal management system is not allowed to delete any mandatory function; system can control optional function, provide a set of configurations that is loaded into the terminal. To ensure inter-operability, the majority of optional card functionality are mandatory for terminals.

The following is an IC card terminal modular structural illustration

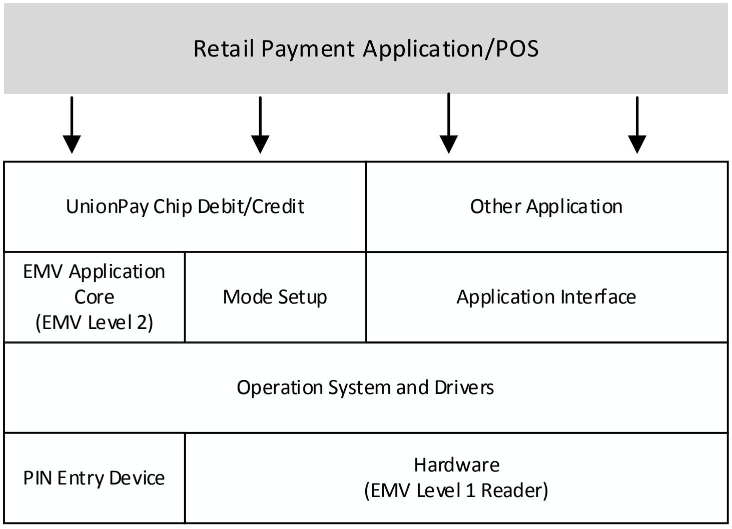

Figure 1 Modularized Architecture

5.2 Manufacturer Selection Standards

Acquirers should emphasis researching the fundamentals of manufacturers, such as registered capital, product scope, org structure, R&D, production, tech support, and post-sale service capability.

- Acquirer should prioritize choice for partnership with manufacturers that have relatively strong capital and technical capability.
- Review at least 2 years of annual financial reports for the enterprise that have been audited by legally qualified registered accounting firm.
- Require the manufacturer produce documentary proof of complete quality control system.
- Acquirer should give priority to manufacturers with a wide selection of product lines.
- Acquirer should give priority to manufacturers with relatively strong financial and industry IC card product R&D capability, application development capacity, and complete post-sale service network.
- Acquirer should give priority to manufacturers with relatively abundant successful implementation experience.

5.3 UnionPay Requirements

UnionPay gives the following recommendation, so as to help acquirers set terminal selection criteria. These recommendations include basic functionality, recommended functionality, and other considerations.

5.3.1 Basic Terminal Functionality

Acquirer must ensure that terminals support the following basic functionality.

Table 1 Basic Terminal Function

Function	Description
Magstripe read capability	Terminal must be able to read the complete information on tracks 1, 2 and 3, and Hi-Co and Low-Co magstripe. During migration, the pure magstripe card will exist for quite a long time, so these requirements are necessary. Keep the magstripe on the chip card, so that it can be a backup if the chip cannot be read.
Contact IC card read capability	Terminal must be able to read contact chip cards and must pass UICS/EMV Level 1 and Level 2 test certification.
Service code identification capability	When processing magstripe, must read expanded service code and process accordingly: • EMV chip presence: 2xx or 6xx
PIN entry	Based on cardholder verification parameters encoded on the chip, the terminal must be able to prompt and accept PIN. Based on issuer selection, requirements of PIN input can be targeted at specific transaction type (e.g.: ATM transaction).
Logical interface between chip reader-writer and magstripe reader-writer	After identifying a chip reading failure, terminal can judge that this card initiates next transaction as magstripe card and accordingly set up corresponding status (through field 60.2.3) and request online processing.
Static data authentication (SDA)	SDA is a public key security verification method, used for verifying that card data is created by a valid issuer, and static data in the card is not modified. It provides an offline chip risk management function, and can protect against counterfeit by verifying critical data at point of transaction. SDA is mandatory to all the terminals offline-capable, but is optional to terminal online-only
Dynamic data authentication (DDA)	DDA is a chip risk management method based on public key mechanism for offline environment. It can protect against "cloning" of data through verifying dynamic digital signature generated at point of transaction. DDA is mandatory to all the terminals offline capable, but is optional to online-only terminals.

Root CA public key loading	To perform offline data authentication or process offline enciphered PIN, terminal must load root CA public key. Terminal must have the capability of loading new public key and removing expired public key.
Variable length account number acceptance	Terminal shall be able to accept account number between 13 and 19 digits.
Point of service entry	When the “point of service entry” mode is set to chip, all the transaction data must come from the chip. For transaction initiated by chip, terminal shall use “magstripe 2 equivalent data” other than track data in the magstripe.

5.3.2 Terminal Function Recommendation

Acquirers can determine whether to support these recommended functions based on their own requirements.

Table 2 Recommended Terminal Function

Function	Description
Offline PIN	All terminals should consider supporting offline cleartext PIN. Terminals able to succeed in verifying offline PIN will bring more offline transactions.
Cardholder application selection	Multi-application is a critical feature of chip card, so terminals should be able to provide this functionality. When card and terminal both support this function, the terminal can display different applications through interaction between terminal and card for cardholder to confirm selections. The terminal shall support at least 2x16 display, and 4 x 20 with bitmap graphic support is highly recommended.
Capable to apply different floor limit to chip and non-chip transaction	As chip card’s security level is obviously higher than magnstripe card, acquirer can consider implementing different floor limit to chip and non-chip transactions. It is recommended that terminal shall be able to provide separated floor limit.
Online capability	Terminal can be offline only, online only, or online-and-offline capable. If terminal does not support online, it can only decline transaction when offline counter reaches its maximum limit; therefore, it is strongly recommended that all terminals shall support online. So hardware must support telecommunication, such has modem or ethernet module, or GPRS/CDMA/3G

	communications module, etc.
Online PIN support	The UPI acceptance environment requires online PIN generally, so all ATM must support online PIN.
Support EC function	For UICS card with EC function, this optional function allows terminal to perform quick offline transaction.
Additional application	Capability to process other applications, e.g.: loyalty program.

5.3.3 Other Considerations

The following descriptions are useful for acquirer to consider terminal design option.

Table 3 Design Option

Function	Descriptions
Processor	The time for the terminal to complete transaction must not be too long for cardholder and cashiers to wait excessively. Acquirer must work with terminal vendors, to ensure their products' processing speed can meet customers' service requirements.
IC card telecommunication speed	To maintain a reasonable response time, the telecommunication speed must reach at least 9600bps.
Memory	The terminal shall have enough memory to process all the applications, support longer key and reserve space for supporting expected applications. Memory shall support firewall isolation so that to ensure cross-application security, and avoid mutual interference.
Modularized Software architecture	Multi-application processing requires creating modularization of code into core libraries, so that all applications can use them. The following modules are recommended: ● Cardholder and cashier's interaction interface, e.g.: table-driven prompters and responses ; ● Drivers for peripherals, e.g.: integrated POS connection and printer interface; ● Communication message processing program. Card shall adopt modularized software architecture so that

	core/software which has passed UICS authentication is not affected.
Flexible architecture	When constructing terminal software and management systems, acquirer shall keep a flexible architecture to avoid major changes which even may influence terminal’s infrastructure in changing information in the future.
Download capability	Updating application, replacing application, loading new application, adding or removing key etc., all require terminal to have download capability. Furthermore, download for updates shall be done at library level, avoiding completely re-download terminal’s device code for a minor coding change.
Combined DDA/ application cryptogram generation (CDA)	CDA combines dynamic data authentication (DDA) and application cryptogram generation, providing authentication for data in the card, card itself and transaction security. CDA is optional to all terminals.
Miscellaneous requirements	Terminal must provide the following components: ● Key pad to enter transaction amount, select command and perform functions; ● Printer used to print transaction slip, choose stylus printer or thermal printer according to payment system requirement; ● Clock for terminal offline-capable so that it can provide local date and time; it is recommended to automatically synchronize this clock to acquirer’s host.
Software version	Terminal IC card software module and application software should automatically determine unique versions at source code compile time and include into acquirer software version management.
TMS	It is recommended that the acquirer should require the terminal to support remote download management capability.

6 Standard Debit/Credit Terminal Requirements

This section will help acquirers understand the requirements for deploying IC card terminals, including requirements of security, interoperability, and functionality described in the IC card specification. It will explain every application function required by terminals, the required policy, operation and technical activities for realizing these functions.

The table below describes the application function covered in this section:

Table 4 UICS Application Function

Function	Description
Application selection	Terminal generates a candidate application list supported by both the card and terminal, and decides to select the appropriate application to complete the transaction.
Offline data authentication	Terminal verifies the card to prevent data manipulation and skimming.
Processing restrictions	Terminal checks whether the application transaction is allowed to continue.
Cardholder verification	It performs appropriate mechanism to verify cardholder.
Terminal risk management	Terminal checks the floor limit, exception file (if exists) and random selection transaction requiring online processing, etc.
Terminal action analysis	Terminal decides how to continue the transaction (approve offline, decline offline or send online) based on the results of offline data authorization, processing restrictions, cardholder verification and terminal risk management and the risk management parameter configuration in terminal and card.

This section only summarizes activities required for realizing each IC card function. For more detailed operation, please refer to EMV and UICS. Acquirers shall communicate with vendors and clearly define their respective responsibilities.

Note: Acquirers shall note that the relevant policies are driven by the overall regional market rather that their policy alone.

6.1 Application Selection

While accepting a standard debit/credit card, the terminal determines which applications are supported by both the card and terminal by comparing the AIDs in the card and in the terminal. The terminal sets up application selection supported by both the terminal and card through directory selection and AID list selection. Directory selection method is mandatory for the terminal but optional for the card. AID list selection method is mandatory for both the card and terminal.

If both the card and terminal support directory selection method, the terminal reads payment application list on the card through a file called Payment System Environment (PSE). The terminal compares the application listed in PSE and that it supports, and sets up a candidate list which is a terminal internal list supported by both the card and terminal. If the card has no PSE, the terminal turns to adopt the AID list selection method.

While adopting the AID list selection method, the terminal enquires one by one whether one application is included in the card application list based on the terminal application list. The terminal adds the application supported by both the card and terminal to the candidate list.

6.1.1 Directory Selection and AID List Selection Methods

UnionPay has designed two different methods to realize application selection:

- Directory Selection Method

This method is mandatory for terminals but optional for cards. Terminals shall apply this method first to read payment system environment (PSE) file from cards. For cards supporting directory selection method, PSE is a high-hierarchy directory file, at least including all the IC card payment application lists in the cards. The terminal compares this list with the list it maintains and adds the application supported jointly by the card and terminal to the candidate list it sets up. Then the terminal selects application according to whether it supports the cardholder selection or terminal automatic selection.

- AID List Selection Method

This method is mandatory for both terminals and cards. While applying AID list selection, the terminal sends a command to the card for every application it supports, enquiring whether the card also supports this application. If the card responds that this application is included in the card, the terminal adds this application to the candidate list. Then the terminal selects application according to whether it supports the cardholder selection or terminal automatic selection application.

As the directory selection method is optional for cards, when cards do not support this method, switch to the AID list selection method. After the terminal receives the candidate application list supported jointly by the card and terminal, one of the following applications is performed:

- Cardholder Selection Application

If the terminal supports the cardholder selection application, any method below can be adopted:

  - ✓ Cardholder Selection

The terminal will offer an application list to the cardholder for selection according to priority. The cardholder chooses an application from the list.

  - ✓ Cardholder Confirmation

The terminal first offers the application with the highest priority to the cardholder for confirmation; if the cardholder does not confirm, the terminal offers the next application with the highest priority until the cardholder confirms or no more application exists.

- Terminal Automatic Selection Application

If the terminal does not support the cardholder selection or confirmation, it will automatically select an application with the highest priority and not require confirmation. If there is more than one application with the highest priority, the terminal can select any of them or the application on the top of the list according to the order that the terminal lists.

6.1.2 Application Identifier

The terminal must include the application identifier (AID) of all the chip payment applications it supports. If the terminal cannot match against the AID of the card with any of the AID it supports, the terminal will not be able to perform chip transaction. AID is composed of two components:

- Registered Application Provider Identifier (RID)
- Proprietary Application Identifier Extension (PIX)

For RID payment plan, UnionPay RID is A000000333; PIX represents applications. For UnionPay PIXs, please see the table below.

Table 5 Proprietary Application Identifier Extension of UnionPay (PIXs)

Application	**PIX**
Debit/credit application	0101
Debit application	010101
Credit application	010102
Quasi-credit application	010103
Pure electronic cash application	010106

The terminal sets an application selection indicator (ASI) for every application, it supports to indicate whether this application is full name match or partial name match. It is recommended that acquirers set ASI to support partial name match UnionPay's AIDs.

For decision-makers, operation personnel and technicians, activities related to AIDs include:

- Policy

Decide what AIDs to support. The terminal shall support the corresponding AIDs of the card brand that the acquirer intends to accept. For example: if acquirer intends to accept UnionPay credit card, terminal shall support UnionPay AID - A000000333010102.

- Operation

Obtain AIDs list that the terminal requires, and work with technicians to load AIDs into the terminal together with a technician and conduct testing to ensure normal operation.

- Technology

Be responsible for uploading AIDs to the terminal and providing the mechanism for uploading AIDs and applications to the terminal.

6.1.3 Language

Acquirers shall decide to offer which language for customers. IC card terminals can display information in several languages and shall support at least one language. For decision-makers, operation personnel and technicians, activities related to terminal-supported languages include:

- Policy

Determine which languages that the terminal can display. Acquirers shall at a minimum support the local language. In addition, if there are a large amount of customers from specific countries, acquirer can consider supporting languages they use.

- Operation

Introduce languages that the terminal can use in the operation and training manual.

- Technology

Work with the terminal vendor, ensuring that the terminal can support languages required.

6.1.4 Cardholder Selection/Confirmation

UPI strongly suggests that the IC card terminal shall support the cardholder selection so that the cardholder can select the application that he wants for the transaction when the card supports multiple applications.

When the terminal and card support more than one application, the terminal can prompt the cardholder selection by any of the following methods:

The terminal displays all the applications supported jointly by the card and terminal in priority sequence preferred by the issuer for the cardholder's selection;

The terminal displays all the applications one by one supported jointly by the card and terminal together in priority sequence preferred by the issuer for the cardholder's confirmation;

- When the terminal supports the cardholder selection and only one application is available:

- If the application requires the cardholder confirmation, the terminal must display this application for the cardholder to confirm whether conduct this application or not (no confirmation will result in transaction termination);

If the application does not require the cardholder confirmation, the terminal automatically selects this application. It is suggested that the terminal display this application to inform customers, but can continue the transaction without waiting for the cardholder confirmation.

- In some special acceptance locations (for example, toll roads, fast-food restaurants), the acquirer can decide not to support the cardholder selection. The acquirer shall aware that some issuers may require the cardholder to confirm the application compulsorily. In these acceptance locations, the cards will not be accepted.

- Activities related to the cardholder selection are as below:

Policy

Decide whether the acquirer supports the cardholder selection. For some special locations, the acquirer can decide not to execute the cardholder selection. If condition permits, it is suggested to conduct the cardholder selection in order to avoid transaction failure due to no customer selection.

- Operation

Introduce the cardholder selection in the operation and training manual.

- Technology

Work with terminal vendors to ensure that the terminal can meet the acquirer' requirements

6.2 Offline Data Authentication

- This Specification defines three offline data authentication methods: Static Data Authentication (SDA), Dynamic Data Authentication (DDA) and Combined Data Authentication (CDA). All these methods apply the public key technology to protect against data counterfeit. However, DDA is more sophisticated than SDA as it generates a dynamic signature for every transaction to protect against counterfeiting. SDA is comparable to the CVN check of magstripe transaction. Both of them can detect whether special static data has been manipulated after personalization. But the difference between SDA and CVN check is that: it is performed at the terminal offline and safer than CVN. CDA processing provides DAA protection and can also ensure that transaction data is not manipulated in the transmitting process of card and terminal.

At present, SDA and DDA are mandatory for offline-capable terminals (offline only or online/offline). Online only terminals don't need to support this function. Therefore, these functions are optional for the online ATM and other merchant environment where the floor limit is set as zero.

Except for online-only terminals, DDA is mandatory for all other terminals.

Note: Before loading production key, the acquirer shall remove all the testing keys to avoid offline data authentication failure.

Activities related to offline data authentication include:

- Policy

Decide whether to support offline data authentication, if so, which method to adopt. Generally, whether to implement SDA, DDA and CDA (or their combination) is driven by the mandatory regulations of the country or region. UnionPay requires that all the terminals must support DDA; otherwise, transactions can only be processed online. UPI suggests that acquirers consult with every participant to decide which offline data authentication method shall be adopted; this requires the acquirers understand the market drivers for migration to chip, and weigh costs and benefits of alternatives.

As migration to chip is inevitable, to save upgrading cost, it is recommended that IC card terminals support offline data authentication function from the beginning.

- Operation

Work with vendors to implement selected offline data authentication method.

- Technology

Vendors shall accomplish relevant development work; generally, in-house technical resources are not likely needed.

> Note: Due to risk control considerations, it is not recommended for terminals to only support SDA.

6.3 Processing Restrictions

Processing restrictions conducted by the terminal check whether the transaction is allowed to continue. It checks whether the card (application) is valid or not, the application version numbers of the card and that of terminal are matched or not, and any restriction condition of application usage control (AUC) is effective or not. Issuers can use AUC to restrict whether cards can be used locally or abroad, for cash withdrawal, goods or service purchase or cash back.

Processing restrictions is a mandatory for IC cards. All terminals are required to support the processing restriction function.

Activities related to processing constrains include:

Policy

No decision needed.

- Operation

Determine the application version number supported by terminals; load it to terminals together with other payment application version numbers supported by terminals.

- Technology

Load appropriate application version number to terminals; when application version number upgrades, load new application version number to terminals.

6.4 Cardholder Verification

- Cardholder verification is used to ensure cardholder is valid and the card is not stolen. The terminal applies the cardholder verification method (CVM) list provided by the card to decide the method to perform verification. For example, signature or offline plaintext PIN. CVM list establishes the priority of the cardholder authentication method and prompts users to adopt the specific cardholder authentication method according to the terminal capability and transaction features. If the card does not support CVM processing, the terminal shall process according to the verification method in the debit/credit application of the magstripe card. If the CVM is "CVM not required", the terminal shall process according to the default cardholder verification method for this type of terminal as specified in specifications. For example, if the default method is signature, the terminal shall print slip with signature line. Some terminal types (for example: ATM) shall always support online PIN entry, no matter whether it is supported in CVM list.

Note: ATM does not allow offline PIN.

UPI recommends merchants to keep CVM information for at least 180 days for dispute resolution. CVM information is corresponding to the selected CVM, for example, keeping transaction slip to prove the terminal has performed offline clear text PIN verification.

CVM supported by This Specification are listed in the table below:

Table 6 Cardholder Verification Methods

Cardholder Verification Method	**Description**
Signature	The operation of this method is in the same manner as magstripe environment: cardholder signs on the sales slip; merchant compares it with the signature on the card.
Offline plaintext PIN	This is a new method: terminal prompts cardholder to enter PIN and sends it to card through plaintext; card compares it with the offline PIN stored in card.
Offline plaintext PIN and signature	The combination of offline plaintext PIN and signature. The two methods are used to perform cardholder verification.
Online PIN	The operation of this method is in the same manner as magstripe environment: terminal applies DES technology to encrypt the PIN that cardholder enters, and

	sends online request to issuer for verification.
CVM not required	This method operates in the same manner as in the magstripe environment where transaction authorization is independent of cardholder verification. No cardholder verification is currently allowed in some merchant environments, such as certain cardholder-activated terminals.
CVM failure	This method allows issuer to default CVM processing as failure.
ID	Terminal prompts cardholder to present ID and displays the card type and number in the card to cashier for cardholder ID comparison.

Support for more secure electronic CVMs with chip transactions, such as Offline PIN, will provide a significant incentive for issuers to migrate to chip and request offline PIN verification. Moreover, the use of offline PIN at chip-reading terminals typically provides additional acquirer liability protection in the event of disputed transactions.

UnionPay mandatorily requires terminals to support cardholder verification and decide the verification method that transactions shall adopt through checking card CVM list (if exists).

Which CVMs the acquirer’s terminals are required to support shall depend on comprehensive consideration of commercial demand, card type to accept, market strategy and UPI requirements. The general rules listed below can help the acquirer decide which CVM its terminals support:

ATMs and other self-service terminals must support "online PIN";

Attended terminals must support "offline plaintext PIN", "online PIN", and continue to support signing methods.

- Offline-only terminals can opt not to support CVM, for specifics please see relevant UnionPay business rules.
- In some markets, for POS transactions exceeding a certain amount, in addition cardholder signature, it probably also requires cardholder to enter offline PIN;

6.4.1 Unknown CVM

- Terminals must be able to identify CVM defined by terminal, even though it does not support this kind of CVM. For example, the CVM list of an IC card includes offline plaintext PIN, signature and “CVM not required,” and the terminal can support signature and “CVM not required” but does not support offline plaintext PIN. In this circumstance, the terminal must be able to indentify “offline plaintext PIN” method to avoid triggering the condition for “unknown CVM.”

6.4.2 Implementation activities

Activities related to cardholder verification include:

Policy

- Decide which CVMs the terminal supports based on UnionPay, local market requirements and commercial consideration;

- Operation

Work with terminal vendors to ensure terminals to support CVM required/recommended/optional requirements for the market where the acquirer is located;

- Technology

The technical department shall participate in the relevant analyzing work, though development work is usually performed by the terminal vendors.

- For online PIN verification, requirements of UICS shall be in consistent with the current regulations.

6.5 Terminal Risk Management

Terminal risk management is mandatory for terminal controlled by merchants, it verifies:

Whether the transaction exceeds the floor limit for the merchant;

Whether the account number of the card appears in the exception file;

- Whether it exceeds the limit of consecutive offline transactions;
- Whether the card is a new one;
- Whether the merchant requires online authorization;
- Whether the transaction is randomly selected for online processing.
- Some functions are probably not suitable for terminals offline only or online only.
- Terminal risk management requirements:

Floor limit of terminal: mandatory for all terminals;

Random transaction selection: mandatory for terminals online /offline;

- Frequency check: mandatory for online terminals supporting offline transactions;
- New card check: mandatory for all terminals.

6.5.1 Floor limit

- Specification mandates that offline-capable terminals must perform the floor limit check for all the transactions.

- Like magstripe terminals, IC card compatible terminals must support the floor limit function. Through card and cardholder verification, chip processing can provide greater security for the transaction. For some markets, chip floor limit may be set as different from magstripe floor limit. If the card parameter identities that transaction must be processed online, the terminal will request online authorization regardless of the floor limit.

In order to provide the maximum flexibility, UPI suggests setting up terminals to support the following floor limit, even if the values for the magstripe and the chip are the same currently:

floor limit of the international magstripe transaction

floor limit of the international chip transaction

- floor limit of the domestic magstripe transaction
- floor limit of the domestic chip transaction
- Note: Zero floor limit is always used for fallback transaction.
- Supporting these independent floor limits in the terminal and its back-office system can provide greater flexibility for the acquirer to meet the future requirements. UPI recommends store the floor limit information based on AID, because the value of floor limit for different card products will change accordingly.

Activities related to the floor limit include:

- Policy

Decide whether there is any new requirement on the floor limit.

Require offline-capable terminals to perform the floor limit check for all transactions. When the transaction amount exceeds the floor limit, offline approval shall be declined as it may bring risk.

  - ✓ Similar to magstripe terminals, IC card compatible terminals must support the floor limit function. As through card and cardholder verification, chip processing can provide greater security for the transaction, and some markets can set up chip floor limit different from magstripe floor limit: maintaining the current floor limit for magstripe transaction and using higher floor limit for chip transaction.

- Operation

  - ✓ Introduce relevant contents about the floor limit in the operation and merchant training documents.

- Technology

Load appropriate IC card floor limit to the terminal and download updated parameter when necessary. Even though the market does not require applying different floor limits, acquirer' terminals shall also be able to support different floor limits for chip and magstripe transactions.

6.5.2 Random Selection

Random selection is a processing method where a terminal randomly selects a transaction with an amount less than the floor limit to request online. It does not depend on card to control parameters. This function is aimed to prevent perpetrators from capturing terminal action patterns and using counterfeit card to accomplish consecutive low value payment transactions. Random selection is mandatory for terminals with online/offline capability.

Note: As e-cash (EC) function is used specially in offline environment, when terminal processes low value payment transaction through EC parameter, it does not need to perform random selection.

Randomly selection’s request online is based on the following three parameters:

- Threshold amount of random selection: value adopted shall be between 0 and the floor limit of the terminal;
- Targeting percentage of random selection: value adopted shall be between 0 and 99.
- Maximum targeting percentage of random selection: value adopted shall be between the targeting percentage of random selection and 99.

For guidelines about setting up random selection parameters, please contact UPI representatives. Activities related to random selection include:

- Policy

  Decide the random selection parameters of online/offline terminal together with participants of IC card plan participants and representatives of UPI in your market.

- Operation

  Introduce relevant contents about random selection in training documents of operation and merchant.

- Technology

  Load random selection parameter to terminals. Vendors of terminals may load this information to terminals in advance.

6.5.3 Frequency Check

Card personalization parameter can control the number of allowed consecutive offline transactions before requesting online. This processing is always called frequency check. Online/offline terminals must support frequency check.

6.5.4 Exception Files

If a terminal exception file exists, the terminal checks whether the card number appears in the exception file list.

This function is optional.

- ✓ Activities related to exception files include:

- Policy

  Judge whether the terminal needs to support this exception file; if the terminal currently supports exception file, it at a minimum shall continue to support chip domestic and magstripe transaction

- Operation

  Terminal needs to be able to download and upgraded exception files .

- Technology

  Work with terminal vendors to ensure terminal programs able to support exception file processing.

6.6 Terminal Action Analysis

Terminal action analysis is mandatory. The terminal shall decide how to precede transaction (approve offline, decline offline or send online) based on results of offline data authentication, processing restrictions, cardholder verification and terminal risk management, together with the risk management parameters set in the terminal and card. The rules stored in the card are called Issuer Action Code (IAC). It can be sent to the terminal through the card. Payment system rules stored in the terminal is called Terminal Action Code (TAC).

Activities related to terminal action analysis include:

- Policy

No decision needed.

- Operation

Find out terminal action codes suitable for your market from representatives of UPI; introduce relevant contents of TAC in training documents for operation and merchant; if the terminal supports EC function, it can require independent TACs.

- Technology

UnionPay TACs and TACs of other payment system must be loaded to terminals. Terminal vendors shall be able to load this information to terminals in advance. In addition, the terminal management system shall be able to trace TACs configuration and upgrade those configurations by uploading when necessary.

Terminals can support EC transactions through a whole set of independent TACs configuration.

6.7 Other Requirements

If acquirer requires support for contactless standard debit/credit applications, relevant terminals must pass contactless UCIS LEVEL2 Standard
Debit/Credit Application Certification. For contactless transaction, terminals that use directory selection must select applications through
PPSE file access.

7 Other Terminal Requirements

Aside from IC card functional configuration requirements, there are also some mandatory or optional features suitable for terminal devices which may be of great help to the market or business. This section will discuss these configuration requirements:

- Magstripe terminal requirements
- EC function (optional)
- Contactless IC card payment (optional)
- Contactless low-value payment extended application payment (optional)
- Dual currency electronic cash payment
- Proprietary requirements
- Transaction type requirements
- Other application requirements
- Transaction receipt requirements

7.1 Magstripe Terminal Requirements

Current regulations on magstripe card transactions still apply to magstripe card terminals. They also apply to magstripe card transactions performed at IC card terminals. Therefore, acquirers shall implement fallback rules and processing procedures at the point of transaction to support the new chip service code.

7.1.1 Fallback

IC card terminals must be able to accept both chip cards and magstripe cards. Therefore, it has chip card reader and magnetic card reader. When a chip card is inserted to the device, the device will read information with the chip card reader other than the magnetic card reader.

However, in some circumstances, chip cards may not be able to be used in the chip card device, for example, if the chip card or the reader in the chip card device is broken. In such circumstance, the chip card will use its magstripe for the transaction. We call this transaction as a “fallback” transaction. As the security management previously controlled by chip is now implemented by magstripe this transaction is considered as with relatively low security.

7.1.1.1 Fallback Policy

In practice, the acquirer must determine how to handle fallback transactions and the specific steps when fallback occurs.

It is specified in the UPI Operating Regulations that the terminal shall try the chip transaction first. Only when the chip or chip reader cannot work normally, a fallback transaction can be adopted.

Markets that recently implemented chip migration shall comply with this regulation. Markets with more experiences in chip implementations can create stricter policy to avoid occurrences of some types of fallback transactions.

7.1.1.2 Prevention and Processing of Fallback

IC card terminals shall have software to ensure that fallback will not occur when the chip card reader can work properly. When a chip card is swiped across the magnetic card reader, the terminal shall prompt the cardholder and merchant to use chip card reader.

This can be accomplished by using a special chip service code (2xx or 6xx) in the magstripe of the chip card. When the chip service code is read by the magstripe reader, the terminal will not proceed the transaction and shall prompt information. This can ensure that the chip card transaction handled at the EMV/UICS terminal is actually accomplished by chip.

In the *Technical Specifications on Bankcard Interoperability (Version 2.1),* a Fallback transaction can be identified through the combination of the following information:

- If "Point of Service Entry Mode (PAN entry mode)" is not 05, 07, 95, 96, 97, 98(field 22), it indicates that it is a non-chip card reader mode.
- If "terminal reader capability" is 5 or 6 (field 60.2.2), it indicates that the terminal is able to read chip cards.
- If the first number of "service code" is 1 or 2 (field 60.2.3), it indicates that there is a chip in the card.

Although the above prevention methods of fallback is not mandatory, it is strongly recommended, because this will become a mandatory function of the terminal device.

Acquirers shall establish a mechanism for magstripe readers to accept chip cards when chip cards or chip readers fail.

Preferable method is that if the number of reading chip card failures exceeds the prescribed times, the terminal starts up the magstripe reader and requests online authorization by the issuer. If online is not supported, the merchant can obtain authorization manually by call referral. If the magstripe card is damaged or the magstripe reader fails to work, the method of entering the account number manually can be adopted to continue the transaction.

If a fallback transaction does not use the magstripe, for example, a voice transaction, the merchant shall clearly indicate that this is a fallback transaction, so that the issuer can take appropriate actions.

7.1.1.3 Implementation Activities

Activities related to fallback include:

- Policy

  Comply with UPI Operating Regulations. Establish domestic policy together with UPI and participants in your market. Train merchants on fallback operation. For more details, please refer to relevant sections in “merchant training”

- Operation

  Cooperate with technicians and terminal vendors to ensure that terminals can prompt merchants to read chip card information through chip readers.

- Technology

Confirm that terminals can be modified to be able to prompt merchants to use chips for transactions. Implementing fallback policy need to add code in terminals. If the terminal supports fallback, it shall support the logical conversion between the magstripe reader and chip reader so that when the chip reading fails, the fallback can proceed the transaction.

For information about terminals, interface of acquirers and modification information of host systems of acquirers involved in the identification and accomplishment of fallback transactions, please refer to Section 11 Acquirer System Modification.

7.1.2 Service Code

IC card introduces new service code values. The first number of the service code set in IC cards must be one of the following two values:

- 2 - Used Internationally

  Indicate that the card has a chip. If the chip works normally, the financial transaction shall be processed by the chip.

- 6 - Used Domestically

  Indicate that the card has a chip. If the chip works normally, the financial transaction shall be processed by the chip.

The current IC card rules require that all online terminals can perform transactions through reading service code. Acquirers must ensure that the new service code will not be declined in the terminal only supporting magstripe. This is not a new rule. If the existing terminal does not comply with this requirement, the program in it should be upgraded.

7.1.3 Implementation activities

Activities required in supporting the chip service code at the magstripe terminal include:

- Policy

  Acquirers must ensure to support the same processing requirements of all the service codes for chips as those for magstripes.

- Operation

  Acquirers shall investigate and analyze the existing terminals based on understanding the influence of new service code values. Acquirers can issue a self-test tool to merchants or arrange a working team to merchants' sites for on-site test. In order to assist the terminal detection, UPI can provide test cards with new service code. If there is compatibility problem in deployed terminals, the operation department shall establish and implement a modification plan together with technicians and terminal vendors.

- Technology

  If necessary, upgrade the terminal software to support the new service code, and test the relevant functions.

7.2 EC Function

Electronic Cash (EC) is an optional function of IC cards. Through this function, issuers can recruit merchants that mainly perform low value payment cash transactions, thus expand IC cards acceptance and provide more applications and convenience for cardholders. Performed offline, an EC transaction does not require any cost of online authorization. This advantage and the fast transaction can facilitate merchants that traditionally only accept (low value payment) cash transactions to shift to accept card transactions.

7.2.1 Requirements Summary

Terminals supporting EC function must be IC card compatible terminals. On the basis of standard UICS, acquirers only need small changes to support EC function. Changes involved in EC terminals are described as below:

- Must acknowledge that the card initiates an EC transaction request, and send an appropriate response to the card;
- It is recommended that the terminal shall contain independent EC TACs which are used to handle EC transactions;
- Must be able to identify the indicator of an EC transaction – the authorization code of EC issuer;

- Terminal risk management of EC transactions shall not perform random selection;
- Can maintain an independent EC floor limit (optional);
- Can display or print the EC available balance on the transaction receipt (optional).

Acquirers can consider enhancing its back-office system to separate EC transactions from the standard IC card transactions for convenience of statistics and accounting.

7.2.2 Data Element Involved

EC application is developed based on UPI Corporate Standards. The card application is approximately similar to the common standard debit/credit card application. However, for easy use, some data elements not included in standard application are added into EC application, the detailed terminal data included can be found in the table below:

Table 7 Newly Added Proprietary Data Elements of EC

Data Element Name	**Requirements**	**Description**
EC Terminal Support Indicator	Condition If supports EC	It is used to indicate that the terminal supports EC function.
EC Terminal Transaction Limit	Optional	The terminal uses this data element (if exists) to judge whether a transaction is handled by means of EC.

To realize electronic cash functionality, the usage of original standard data elements in EC function will also be different. Some data elements are mandatory, for example, transaction currency code. For detailed terminal data element impacted by EC, refer to the table below:

Table 8 Data Elements Impacted by EEC

Data Element Name	**Requirements**	**Description**
Authorized amount	Condition If supports EC	If the terminal supports EC function, card will require terminal to provide the data element of authorized amount through PDOL. This element is used for card to judge whether the transaction

		meets the EC transaction condition.
Transaction currency code	Condition If supports EC	If the terminal supports EC function, card will require terminal to provide data element of transaction currency code through PDOL. This element is used for card to judge whether the currency is the same.

7.2.3 Transaction Process of EC

Interaction between an EC-capable terminal and a card with EC function includes the following steps:

1. During application selection, the card with EC function will return PDOL, requesting the EC terminal to support the indicator, transaction amount and transaction currency code.

2. The terminal judges whether it supports EC, transaction amount and other factors And Set up the EC terminal to support the indicator, provide PDOL requesting data and send GPO command to the card.

3. If the card judges that the indicator supported by the EC terminal is 1, it perform the following checks:

   - ✓ Is the transaction currency code equals to the application currency code?
   - ✓ Is the transaction amount equal to or less than the EC available balance?
   - ✓ Is the transaction amount equal to or less than the single EC transaction limit (if exists)?
   - ✓ Is the issuer authentication of the previous online transaction successful?
   - ✓ Does the offline PIN try exceed the limit?

4. If all the conditions above are positive, the card identifies this transaction as an EC transation, then returns the EC AIP and AFL to the terminal.

5. If any of the condition above is not accorded with, then the card returns the EC AIP and AFL to the terminal.

6. The terminal reads the corresponding application record, if the terminal finds that the EC issuer authorization code exists, use the EC TACs for the follow-up processing, otherwise, use standard TACs. The follow-up processing is performed according to the standard procedure.

7. During terminal action analysis, for an EC transaction, in addition performing the original processing logic, the terminal needs also to check:

   - ✓ Whether the EC balance is less than the threshold of EC reset value?

- ✓ Whether the terminal supports online?

The terminal comprehensively decides the type of the cryptogram requested: TC, ARQC or AAC.

8. During the period of card action analysis, if the card judges that it is an EC transaction, it is processed as follows according to the type of the cryptogram requested.

    - ✓ If request TC, deduct transaction amount from EC balance and return to TC;
    - ✓ If request AAC, return to AAC;
    - ✓ If request ARQC, return to ARQC.
    - ✓ For standard application equipped with the EC function, the EC balance shall be returned through the issuer discretionary data (IDD) of the issuer application data (Tag ‘9F10’).

9. The termial continues to handle the transaction according to the standard process.
10. The terminal shall include the EC issuer authorization code in the EC submitted settlement information.

7.2.4 Implementation activities

Activities required for realizing the EC function include:

- Policy

  Adhere to "UPI Operating Regulations" and ensure IC card terminal, card, and back-office systems all meet electronic cash acceptance capability.

- Operation

  Acquirer technical staff should partner with supplier to create and implement an upgrade plan for terminal, card, and system application, ensuring that the merchant can acquire electronic cash.

- Technology

  If necessary, based on the UICS/EMV Level 1 and 2 internal reviews that the terminal has to pass, modify terminal application software and support functionality for electronic cash, and test relevant functionality for accuracy. For terminal application software modifications, please refer to "UnionPay Integrated Circuit Card Specification – Basic Specification Part 3 Debit/Credit Application Terminal Specification ".

7.3 Contactless IC Card Payment

The development of the new contactless technology brings challenges and innovations to the existing financial transaction method and technical environment. Considering cardholders' habit of using card and preventing the transaction from being interrupted, it is critical to shorten the transaction time as much as possible while ensuring the smooth operation of transactions.

7.3.1 Requirements Summary

7.3.1.1 Level 1 Requirements

IC card terminals that comply with contactless specification shall meet Level 1 requirements:

- Terminals shall meet the *ISO 14443 2001*;
- For Type B protocol, terminals shall support MBLI=0 and MBLI=1;
- For Type A cards, terminals shall support the FWI with the value of 8 and ATS–TB (1) with an added value of x 'B'.

7.3.1.2 Basic Requirements

Basic requirements for IC card terminals complying with contactless specifications include:

- The offline-capable terminal shall support both SDA and DDA . If the card supports DDA, the terminal shall perform DDA.
- The terminal shall support data object list (DOL) defined by UPI Corporate Standards.
- The acquirer can provide this information by correctly filling in the mandatory items in the online message –"Point of Service Entry Mode"
- If the card returns the application authentication cryptogram (AAC) to decline the transaction, the transaction shall not be processed through other interfaces.
- When initializing a contact or magnetic swiping card transaction, the terminal shall deactivate and close the contactless interface.
- When initializing the contact or magnetic swiping card transaction, if the contactless transaction is in process, the terminal shall terminate the contactless transaction, abandon all data received from cards and restart other interfaces to perform the transaction.
- For a contactless transaction, the terminal shall inform the cardholder and merchant of the following information:
    - ✓ Present card

  - ✓ Transaction process
  - ✓ Transaction result - approval, decline or terminate

- Recommended terminal information includes:
  - ✓ Presenting card
  - ✓ Card reading succeeds
  - ✓ In processing
  - ✓ Present card again (if the transaction is not finished)
  - ✓ Transaction approved
  - ✓ Transaction declined
  - ✓ Present a single card (preventing collision)
  - ✓ Insert or swipe card
- When prompting to present a card, the terminal shall display the transaction amount (Tag '9F02).
- If the card provides "offline purchase available balance" (Tag '9F5D), the terminal shall display this amount to indicate the successful operation of the card reading and might print the amount on the transaction receipt.

7.3.2 Quick Debit/Credit Payment Application

qUICS is based on standard debit/credit transaction concepts, using existing systems and operating rules. By decreasing the number of commands and responses, qUICS shortens the processing time between the terminal and card. It provides the offline quick low value payment feature, offline data authentication and the online card authentication applying the existing cryptogram algorithm (version 01) or new simplified algorithm (version 17).

In order to meet the requirements on transaction speed for introducing contactless interface, the standard debit/credit application process must be adjusted and enhanced. qUICS enhances the command and transaction process mainly in the following aspects:

- Compress several EMV commands as few as possible to shorten the transaction time;
- Complete the interaction of the card and terminal in combination. When the card leaves the communication range of the card reader, the terminal performs offline data authentication, terminal risk management and terminal action analysis and allows the card to perform the cipher operation before and after leaving the induction area of card reading to make the time of the card in the induction area as short as possible.

The qUICS has two major characteristics:

- Online transaction adopts online card authentication;
- Offline transaction adopts offline data authentication.

7.3.2.1 Requirements Summary

In addition the general requirements for all contactless applications, the terminal supporting qUICS shall also meet the following requirements:

1. The terminal shall support qUICS transaction preprocessing.
2. Terminal supporting qUICS shall read the record according to EMV rules, and process the data element encoded by the unknown Tag in the record or PDOL.
3. If the qUICS mandatory data element is not returned by GPO, the terminal supporting qUICS shall decline the transaction.
4. If the data element required by standard application but not needed in qUICS does not exist, the terminal supporting qUICS shall not decline the transaction.
5. The terminal supporting qUICS shall provide “magstripe 2 equivalent data” in any of the qUICS online message requiring magnetic track data.
6. If the card does not provide “card transaction qualifiers” (Tag ‘9F6C’) data element, the terminal supporting signiture shall suppose that the card requires signiture; if the terminal requires CVM, the signiture bar shall be printed on the transaction receipt.
7. The terminal supporting more than one CVM shall decide which CVM the card will select by inquiring bit 8 and bit 7 of byte 1 of the card transaction qualifier.
    - ✓ If bit 8 = ‘1’, the terminal shall perform online PIN verification and need not inquire bit 7; if bit 8 = ‘0’, the terminal shall inquire bit 7;
    - ✓ Unless the terminal supports online PIN, the card will not configure bit 8; the present card logic will not configure bit 8 and bit 7 simultaneously. However, CVM added in the future may also require the card logic to change;
    - ✓ If bit 7 =’1’, the terminal shall print the signature bar on the receipt.
    - ✓ If the application cryptogram does not present in the GPO response, the terminal shall process as a standard transaction;
    - ✓ If the application cryptogram presents in GPO response, the terminal shall process the transaction as qUICS.
8. In any of the following circumstances, the offline data authentication fails and the transaction is declined.

- ✓ AIP does not indicate supporting offline data authentication (SDA or fDDA);
- ✓ Or terminal supports SDA, but lacks the data required by SDA;
- ✓ Or terminal supports fDDA, but lacks the data required by fDDA.

7.3.2.2 Data Element Involved

qUICS does not require all the EMV mandatory data to be included in the card. Or if included in the card, it is not required to be read out. The following table gives respective explanation to the new qUICS terminal data element and the original terminal data elements impacted by qUICS.

Table 9 qUICS Proprietary Data Elements

Data Element Name	Requirements	Description
File name of PPSE	Mandatory	The terminal selects contactless payment application through the file name of PPSE '2PAY.SYS.DDF01'.
Contactless Floor Lifloor limitmit	Optional	It indicates the floor limit of the contactless payment application supported by terminal.
Contactless Transaction Limit	Optional	If the contactless transaction amount is more than or equal to this amount, the transaction terminates. In this circumstance, other transaction interfaces are allowed to be used to perform the transaction.
Terminal CVM Required Limit	Optional	If the contactless transaction amount is more than or equal to this amount, the terminal requires to perform CVM processing. At present, qUICS supports two kinds of CVM: online PIN and signature.

Terminal Transaction Qualifiers	Mandatory	It indicates the processing ability of the terminal and the requirements for the card.

Note: qUICS can provide quick offline low value payment function. It combines EC features in the qUICS transaction process. For relevant EC terminal data element, refer to “Table 7 - EC Proprietary Data Elements”.

The qUICS does not support CDOL, DDOL or default DDOL of EMV. All terminal data required in card processing is requested in PDOL. Based on the cryptogram types (01 or 17) supported by card and whether it supports offline qUICS transactions, the basic terminal data requested by card will be different.

7.3.2.3 qUICS Transaction Process

Interaction between a qUICS card and a qUICS-capable terminal includes the following steps:

1. Transaction preprocessing

   - ✓ The terminal obtains the transaction amount;
   - ✓ If the transaction amount is more than or equal to the contactless transaction limit (if exists), the terminal will try to use other transaction interface;
   - ✓ If the transaction amount is more than the floor limit of the contactless (if exists, otherwise, use terminal floor limit), the terminal shall indicate “cryptogram requesting online processing” (For an offline-only terminal, the transaction shall be terminated);
   - ✓ If the transaction amount is zero, the terminal shall indicate “cryptogram requesting online processing” (For an offline-only terminal, the transaction shall be terminated);
   - ✓ If the transaction amount is more than or equal to the CVM limit performed by the terminal, indicate “request CVM”;
   - ✓ The terminal requests to present a card.

2. Card Detection Processing

   If several contactless cards have been detected at the same time, the terminal shall prompt that only one card shall be presented.

3. Application Selection

   - ✓ The terminal selects PPSE with “2PAY.SYS.DDF01”;

- ✓ The terminal creates an application list supported by both the terminal and card;
- ✓ The terminal selects an application with the highest priority;
- ✓ The terminal judges whether the PDOL response includes “terminal transaction qualifiers” (Tag ‘9F66’) data element. If it exists, the terminal shall start initializing application , otherwise, the terminal shall try other transaction interfaces.

4. Application Initialization

- ✓ The terminal organizes data required by PDOL and send GPO command to the card;
- ✓ The card receives GPO command and configure protection against interruption;
- ✓ If the card supports qUICS and “terminal transaction qualifiers” indicator supports qUISC, the card performs qUICS transaction processing;
- ✓ The terminal performs qUICS card risk management;
- ✓ According to the result of risk management, the card can require termination, decline, online or approval; for different requirements, the card formats GPO response and returns it to the terminal.

5. If the terminal receives GPO termination response (for example, status code = x’6985’), the terminal terminates the transaction.

6. If the terminal receives correct GPO response, it judges whether the transaction shall online process, or offline decline or approve based on the cryptogram type.

   - ✓ If ARQC is returned, the terminal requests online processing:
     - The terminal shall prompt the cardholder and merchant that the card can be removed and the transaction is requested to be processed by the issuer;
     - The terminal performs CVM required (for example, enter online PIN);
     - The terminal approves or declines the transaction according to the response of the issuer.
   - ✓ If AAC is returned, the terminal offline declines the transaction ;

     Note: The terminal shall not try other interfaces to process.
   - ✓ If TC is returned:
     - The terminal reads additional data using AFL responded by GPO;

- If the application expiry date is earlier than the terminal transaction date, the terminal shall offline decline the transaction;
- If the application PAN presents in the terminal exception file (if exists), the terminal shall offline decline the transaction ;
- The terminal performs offline data authentication (SDA or fDDA). If fails, it continues to judge whether it should process the transaction online, terminate or offline decline the transaction;
- If passing the offline data authentication, the terminal offline approves the transaction.

7.3.2.4 Implementation activities

Activities needed in perform qUICS functions include:

- Policy
  - ✓ Decide the specific capability and risk management requirements of qUICS terminals;
  - ✓ Decide whether to use contactless floor limit, contactless transaction limit and CVM limit performed by terminals
  - ✓ As this data bring to acquirers greater risk management for qUICS transactions, UPI recommends using these data in the terminal.
  - ✓ Assess and decide to select which merchants to provide this service.
- Operation
  - ✓ According to the functions to be realized , the acquirer shall communicate with terminal vendors about data element they need to support;
  - ✓ Decide the specific configuration of the terminal transaction qualifier;
  - ✓ Set up contactless floor limit, contactless transaction limit and CVM limit performed by terminals (if supports).
  - ✓ The acquirer shall fully communicate requirements on qUICS to the vendor. It is quite important to describe requirements to vendors clearly, as the application software of some vendors does not support this function at present.
- Technology

  To realize qUICS, terminals must support some data elements described above, for example, terminal transaction qualifiers.

7.4 Contactless Low-value payment Extended Application

Extended applications are mainly used for some specific low-value payment scenarios, currently including two application modes - fee deduction by segment and offline pre-authorized purchase (only supports low value check):

Fee deduction by segment:

Meant for subway, expressway, parking, and other industries with time, segment, and trip recording requirements. On the basis of the original standard transaction flow, in between the processing of GPO command and READ RECORD command, add an UPDATE CAPP DATA CACHE command to update extended application data.

GPO command fee deduction and UPDATE CAPP DATA CACHE update must ensure simultaneous execution, so that they are executed together as READ RECORD command successfully reads the last record in AFL. In fee deduction by segment mode, issuer can choose to support deposit deduction functionality, and must at time of personalization add two tags - fee deduction by segment limit (DF62) and fee deduction by segment completed amount (DF63).

Offline pre-authorization:

In order to meet offline pre-authorization scenario requirements, on the basis of the original fee deduction by segment transaction, extend CAPP transaction indicator bit and application file, add two types of transaction: offline pre-authorized transaction and offline pre-authorized transaction complete.

For specifics please see "UnionPay Integrated Circuit Card Specification- Product Specifications Part 2 Extended Purchase Specification Based on Contactless Low-value Payment Application ".

7.4.1 Requirement Overview

Terminals that support contactless low-value payment extended application functionality must first be UICS/EMV compatible terminals. On the basis of standard applications, supporting contactless extended application functionality only requires minor modification. The terminal modifications involved are broadly described below:

- Terminal equipment reads card, through reading the extended application proprietary file on the card, determine whether extended application functionality is supported.
- Verify extended application proprietary file, to determine whether the card supports a specific industry application functionality, and checks whether the last transaction completed normally.

- In the transaction process need to record purchase related data, with main transaction flow is basically the same as in the standard qUICS transaction flow.
- After transaction completion, using terminal records of relevant data, calculate actual transaction amount and deduct accordingly.
- After transaction is complete, the terminal makes a standard format offline transaction log record and batch uploads it by end of day, through UnionPay network to the issuer bank, completing final settlement.

7.4.2 Relevant Data Elements

Includes the following two parts:

New additional files:

Extended application defines two types of proprietary files - variable length record file and circular record file, with the former being used to record timestamp, route, or other information for entering payment area and exiting payment area, then use the difference between the two data points to calculate the actual purchase amount and complete deduction transaction.

Circular record file is generally used to record log file, such as transfer data, used as the basis for transfer discounts.

New data elements:

Extended application new data elements DF60/DF61/DF62/DF64, including:

DF60 is the extended transaction (CAPP) indicator bit, showing the extended application transaction type supported by the terminal.

DF61 is the fee deduction by segment application identifier, used to indicate capability for card to support extended application.

DF62 is the electronic cash fee payment by segment limit, used to show the maximum amount that can be deducted in a fee deduction by segment transaction.

DF63 is the electronic cash fee deduction by segment completed amount, used to show the amount already deducted on the card.

7.4.3 Implementation Activities

Activities required to support contactless low-value payment extended application functionality on IC card terminals include:

- Policy
  - ✓ Determine whether the terminal will support contactless low-value payment extended application functionality;

- ✓ Determine which application mode the terminal will support.

- Operation
  - ✓ Cooperate with technical staff and terminal suppliers to ensure terminals support contactless low-value payment extended application functionality.

- Technology
  - ✓ Terminal applications must implement support for two application modes.
  - ✓ Terminal application must support the aforementioned files and data elements.

7.5 Dual Currency Electronic Cash

Terminal can support the following new data elements for dual currency electronic cash application .

Table 10 Data elements used by secondary currency electronic cash

Data element name	Tag	Length	Format
EC Secondary Application Currency Code	DF71	2	n4
EC Secondary Application Balance	DF79	6	n12
EC Secondary Application Balance Limit	DF77	6	n12
EC Secondary Application Single Transaction Limit	DF78	6	n12
EC Secondary Application Reset Threshold)	DF76	6	n12

For specifics please see "UnionPay Integrated Circuit Card Specification - Product Specification Part 6 Dual-currency Low-value Payment Specification Based on Debit/Credit Application ".

7.5.1 Requirement Overview

Terminals that support dual currency electronic cash functionality must first be UICS/EMV compatible terminals. On the basis of standard applications, supporting dual currency electronic cash functionality only requires minor modification. The terminal modifications involved are broadly described below:

- Terminal must support purchase transactions, refund transactions, and load designated account transaction.
- Terminals must support UnionPay IC card electronic cash related offline balance inquiry and offline detail inquiry.
- Terminals must support uploading of offline transactions
- Terminal does not support UnionPay IC card electronic cash withdraw transaction

7.5.2 Implementation Activities

Activities required to implement dual currency electronic cash functionality include:

- Policy
  - ✓ Determine whether terminal supports dual currency electronic cash functionality;
  - ✓ Determine which specific transaction types are supported by the terminal.
- Operation
  - ✓ Cooperate with technical staff and terminal suppliers to ensure terminal support for dual currency electronic cash functionality
- Technology
  - ✓ Various transactions of terminal application described above
  - ✓ Terminal application must prohibit use of UnionPay IC card electronic cash for withdraw transactions

7.6 Proprietary Requirements

If it is required to support the card products of some proprietary brands, the acquirer shall carefully plan how to realize these functions in the framework of IC card compatible terminals. This needs the acquirer to discuss comprehensively with the issuing organization of the relevant proprietary brands.

Additionally, customization of chip system architecture provides an ideal opportunity to revisit the functionality and requirements of terminals and can review the historical concerns with merchants and UPI.

Activities related to proprietary terminal function include:

- Policy

  Determine any proprietary terminal requirements with the issuing organization of the proprietary brands in the market.

- Operation

  Confirm support to the proprietary demands with the technical team.

- Technology

  Upgrade the terminal to realize and test the proprietary program. Terminal vendors can provide support.

Note: The scope of UPI test and host authentication does not involve proprietary functions. Acquirers and terminal vendors are responsible for testing proprietary functions.

7.7 Transaction Type Requirements

Chip card terminals must support transactions of various kinds. For the majority transaction types, the processing rules of chip card transactions and magstripe card transactions are the same. There are differences in specific scenarios. This section will focus on these differences.

7.7.1 Electronic cash designated account loading

Designated account loading transaction fee allocation involves electronic cash issuer, acquirer, and UPI the provider of the inter-bank data connection. Transaction yield allocation uses fixed acquirer fee and UnionPay network service fee formats.

7.7.2 Electronic cash non-designated account loading

Non-designated account loading transaction fee allocation involves electronic cash issuer, outbound transfer org, acquirer who provides the machine and agent service, and UPI the provider of the inter-bank data connection. Transaction yield allocation uses fixed acquirer fee, outbound org transfer fee, and UnionPay network service fee formats.

7.7.3 Cash top-up transaction

Cash top-up transaction fee allocation involves electronic cash issuer, acquirer, and UPI the provider of the inter-bank data connection. Transaction yield allocation uses fixed acquirer fee and UnionPay network service fee formats.

7.7.4 Electronic cash refund

IC card electronic cash refund means cancellation of the current batch or prior batch of offline purchase transaction, supporting partial refund and multiple refund. For electronic cash refund process please see "UPI Operating Regulations”.

7.7.5 Reversal/ Cancellation

If a single-message transaction cannot be accomplished after receiving the approval response from the issuer, a reversal shall be initiated. This is applicable for both magstripe card transactions and chip card transactions. In addition to the existing triggering conditions of magstripe card transactions, there are new triggering conditions for chip card transactions. If the issuer approves the transaction online, but subsequently the card declines the transaction offline, a reversal shall be initiated in the single-message transaction environment.

If the acquirer needs to process a single-message transaction, the terminal vendor must be clearly informed that this new triggering condition requires reversal. No other policy, operation and technical activities apply.

The acquirer shall also consider how to process the reversal of a dual message transaction and restore the “authorized credit” of cardholders.

7.7.6 Refund

Refund caused by commodity return or service cancellation can be classified into two processing methods: online refund and manual refund. Online refund transactions of chip cards apply simplified EMV process. The POS terminal reads the card number, card sequence number, expiry date and other relevant data in a chip card. In this circumstance, only application selection, application initialization, reading application data need performing. Offline data authentication, cardholder verification and terminal risk management are not needed to be performed.

If PDOL requires transaction amount, sending a zero amount to the card is recommended.

7.7.7 Informational Advice

For some exception information or information needing special processing in the chip payment application, informing the issuer is necessary. For example:

IC card information file

The IC card information file formulated by UnionPay System includes information of the following two aspects:

- Recorded offline transactions data based on IC card failure;
- Recorded data of IC cards transactions in which although the card verifies ARPC as wrong, it still approves the transaction.

IC card information file is only used as risk information reference, having no relation with settlement.

Processing result of IC card script

When a IC card transaction includes the issuer's script, the acquirer shall submit the performing results of the issuer's script in the form of advice message.

Activities related to performing informational advice include:

- Policy

  If the acquirer requires notice of special information, please contact representatives of UPI.

- Operation

  Describe requirements related to informational advice in the operation manual.

- Technology

  Cooperate with vendors to ensure that the terminal supports the corresponding functions.

7.8 Other Application Requirements

In addition to standard applications, the acquirer can also support other chip card applications. If the acquirer intends to do so, it must enable the terminal to process these applications correctly.

Activities related to implementing other chip applications include:

- Policy

  Decide which other application types to support, for example, loyalty program or prepaid application.

- Operation

  Describe other applications and requirements in the operation manual.

- Technology

  Cooperate with vendors to ensure that the terminal supports the corresponding functions.

7.9 Requirements for Transaction Receipt

7.9.1 Transaction Flow Backup Requirements

For offline transactions, especially for low-value payment transactions without documentation, terminals should have physical flow backup mechanism to ensure access to offline transaction flow when terminal is not in normal operational state.

7.9.2 Purchase Slip Requirements

The card number printed on transaction receipts, besides adhering to any requirements from local business regulators, should hide all numbers besides the first 6 and last 4 digits of the bank card number (except bank card numbers printed for inbound transfer card in transfer transaction, pre-authorized transaction, and electronic cash transaction), each digit of hidden numbers are replaced by the same special character (* or #).

Used to print some reference data, whether these data needs to be printed are generally related to transaction type. Note data that may appear include:

- Standard debit/credit IC card purchase and pre-authorization transactions, should print AID and ARQC
- Quick debit/credit online applications should print AID and ARQC
- Low-value payment based on standard debit/credit applications should print CSN, AID, TC, TVR, TSI, ATC, UNPR NUM, AIP card balance, TermCap, and IAD including CVR
- Low-value payment designated account loading, non-designated account loading, and cash top-up transactions based on standard debit/credit applications should print AID, post-top-up card balance, ATC, and transfer-in card number for non-designated account loading without masking card number.

For low-value payment offline purchase, number of copies for purchase slip printing should be set for 1. Electronic cash transaction receipt requirements:

- Cross-bank transaction generated digital data or paper receipts can be used as proof for settlement, funds transfer, error adjustment, and dispute resolution.
- Digital data and paper receipts that record transaction data should be retained at least 1 year starting from transaction date, if local laws require a longer retention period, then must extend retention period accordingly.
- For load-type transactions (except for auto-loading), issuer issued transaction response data is the main evidence for confirming transaction.
- For auto-lead transactions, UPI inter-bank connection system does not record transaction data.
- For purchase transactions, TC data and relevant data records for calculating TC in terminal and acquirer can be used by issuer as proof in error and dispute resolution.

8 Acquirer System Modification

8.1 Acquirer IC Card Migration Requirements

IC card requires some new chip data for authorization type, financial type, and settlement type transactions. There new data provides:

- Risk management activity result generated at transaction point between card and terminal, e.g.:
  - ✓ Result of counterfeit card verification through offline data;
  - ✓ Result of cardholder authentication through offline PIN.
- Data integrity and data "clone" protection cryptogram data.
- Dispute resolution protection is effective for reducing retrieval and chargeback requests.

In order to use these valuable data, the acquirer host system must be modified. New chip data fields are sent to the acquirer through the terminal, and transferred by the acquirer to the issuer through UnionPay.

Acquirer IC card migration helps to achieve greater competitive advantage through the chip payment service. These advantages include:

- Allowing the issuer to use offline processing, online card authentication results to make approval or rejection decisions during the transaction.
- Lower fraud
  - ✓ Permit cards to execute issuer authentication;
  - ✓ Provide card number and inter-terminal audit proof;
  - ✓ Protect data integrity.
- Boost customer service
  - ✓ Reduce the number of retrieval, chargeback, and re-presentment requests;
  - ✓ Provide more in-depth transaction observation at the transaction location.
- Leverage supplier system advantages and available resources to support implementation.
- If issuer fundamental structure can also support complete migration, then the overall market can benefit from chip card risk management, customer service, and other areas.
- Can adapt to the domestic or international markets that have required mandatory full migration.

- Incentive mechanism, ex.: For complete data transfer, provide a relatively low interchange rate or liability protection. Fully support high efficiency chip data modification in projects, so as to minimize future modifications.

8.2 Standard IC Card Acceptance Requirements on Transaction Flow

Standard IC card transaction (including electronic cash) process generally includes the following steps:

- Initialize transaction

  Insert chip card into the chip reader, or use magnetic stripe read-writer to swipe card. If swiping, terminal checks the service code in the magnetic stripe to determine whether it is a chip card:

  - ✓ If the service code indicates that it is not a chip card, the terminal will continue processing as per magnetic stripe card transaction;
  - ✓ If service code indicates a chip card, the terminal will prompt the attendant and require use of the chip.

- Application selection

  Terminal will compare the applications supported by the card with the list of its own supported applications:

  - ✓ If the card and terminal do not have commonly supported applications - terminate transaction;
  - ✓ If the card and terminal only have 1 commonly supported application and doesn't require cardholder confirmation - choose this application;
  - ✓ If card and terminal shares multiple supported applications in common, or only shares a single supported application but requires cardholder confirmation - terminal display application list for cardholder selection, or choose based on issuer priority list (regardless of whether cardholder confirmation is required) to complete application selection.

- Application initialization processing

  Terminal notifies card of transaction initiation. Card sends data and risk management information to the terminal for use in the transaction process. In the response, card executes the following steps:

  - ✓ Constraint checking (optional) Card can check whether the current transactional environment is not permitted. If this application does not support this type of transaction, the card directs the terminal to choose another application.
  - ✓ Card sends AIP and AFL to terminal (mandatory)

Based on transaction environment, the card can return different API or AFL, where AFL directs the terminal to read which records from the card. For example: Card can return different AIP and AFL for domestic/international transactions.

✓ Read application data

Terminal reads card data needed for transaction processing, preparing data needed for offline data authentication.

- Offline risk management

  Execute offline transaction checking (if supported by card), determine whether transaction should be offline (approve/decline) or processing online. If the transaction must be processed online, the terminal initiates online transaction request.

  The following offline processing checks could be executed in any desired order:

  ✓ Offline data check Offline data check can execute one of the following three methods:

  - Static data authentication (SDA)
  - Dynamic data authentication (DDA)
  - Combined DDA/ Application cryptogram generation (CDA)

- Considering many factors in combination, generally DDA should be used. After offline data authentication, terminal will record the processing result in TVR; if offline data authentication was not executed, transaction must request online processing.

  ✓ Processing constraints Terminal executes the following checks:

  - Effective date check
  - Expiration date check
  - Application usage control check
  - Application version check

  ✓ Cardholder authentication

  Terminal reads the CVM table in the card and chooses the most appropriate CVM for this transaction. Terminal and merchant authenticates cardholder legality, and that the card has not been reported lost or stolen.

  UnionPay supports the following cardholder authentication methods:

        - CVM not necessary
        - CVM processing failure
        - Offline plaintext PIN
        - Offline plaintext PIN and signature
        - Online PIN
        - Signature
        - Produce personal identification document
    - ✓ Terminal risk management Terminal executes a series of checks based on acquirer risk control characteristics:
        - Lower limit amount check
        - Exception file check
        - Frequency check
        - Random selection The check result is provided to the terminal for action analysis to determine transaction offline approval, offline decline, or request online processing.
- Terminal action analysis Terminal uses card and payment system configured action rules (IAC/TAC) to determine how to continue transaction in terms of offline processing result (TVR), terminal requests a corresponding cryptogram:
    - ✓ Offline approval—Transaction certificate(TC)
    - ✓ Offline decline—Application authentication cryptogram(AAC)
    - ✓ Online request—Authorization request cryptogram(ARQC)
- Card risk management
- Permit card to execute frequency check and other risk management checks on behalf of the issuer, to determine whether to agree with terminal decision. Card can execute the following checks:
    - ✓ New card check
    - ✓ Last transaction check
    - ✓ Frequency check
- Once the card completes risk management check, it will use one of the three cryptograms to respond to the terminal's cryptogram request:
    - ✓ Offline approval—TC
    - ✓ Offline decline—AAC

  - ✓ Online request—ARQC Card can return a cryptogram of a different type than what the terminal requested. If the terminal requests offline approval, the card can return online processing or offline decline; if terminal requests online processing, card can request offline decline; but if terminal requests offline decline, the card can only return offline decline.

- Online processing Through terminal action analysis and card action analysis, card and terminal determine transaction and requests online processing. Card generates a ARQC for online processing and sends it to the terminal, terminal obtains ARQC, original data used by the chip to generate this cryptogram, and the result of offline risk management check, then sends these data online to the acquirer. Acquirers that realize complete migration can package these data according to format rules into transaction reports and send to UnionPay System.
- UnionPay System processes acquirer transaction request All chip data will be transferred to issuer;
- Issuer receives transaction request Issuer requires:
  - ✓ ARQC check (online card authentication)
  - ✓ Analyze CVR
  - ✓ Analyze TVR
  - ✓ Analyze other offline processing results
- After issuer makes transaction decision, send response message to UnionPay System. Response content can include:
  - ✓ Authorization response cryptogram(ARPC)—card executes check (optional)
  - ✓ Issuer script (optional)
- UnionPay System process issuer response

If response message comes from a partially migrated issuer and requests proxy check service, UnionPay System generates ARPC on its behalf and sends it to the acquirer to execute issuer authentication.

- Acquirer process issuer response

  Acquirer will send the response data to the terminal, terminal will send the issuer authorization response result and issuer script to the card for processing, which would involve setting a series of counters.

  Card uses authorization response to determine approval or rejection of transaction:

  - ✓ Card checks ARPC, ensures response is from a valid issuer;
  - ✓ Card authenticates and executes issuer script command (if exists)
- Card generates the final cryptogram: TC for approved transaction, AAC for decline transaction, TC and relevant data for calculation of TC can be used as confirmation basis for transaction.
- Clearing and settlement

For online approval and offline approval transaction, card will generate a final cryptogram - transaction certificate (TC).

  - ✓ If transaction is declined offline, card will generate AAC;
  - ✓ Offline declined transaction detailed information will be sent to the issuer through "IC card data file";
  - ✓ Acquirer submits offline approval transaction to UnionPay System for clearing and settlement.

8.3 Terminal Management System

The original terminal management system needs upgrading and modifying to support standard and other chip card plans. For UPI standard IC cards, the acquirer must be able to process terminal data verification and terminal data upgrading after configuration.

In order to modify the system without causing any major changes to the system structure, the original terminal management system needs to have sufficient stability and flexibility. The more flexible the original terminal management system is, the easier it can be expanded and adapted to future business needs.

Data elements to be verified and updated after terminal deployment usually include:

- Root CA public key
  - ✓ In order to support installation, tracing and withdrawal of root CA public key (for terminals supporting SDA, DDA or CDA), the terminal must possess the capabilities of accessing, adding and removing public keys;
  - ✓ Ensure that the terminal can safely store at most 6 UICS root CA public keys (and related data);

The current effective public key length is between 1024 bit and 1984 bit. For more information related to public key, please refer to the *UnionPay Integrated Circuit Card Specification - Auxiliary Specifications - UnionPay Integrated Circuit Card Debit/Credit Application Root CA Public Key Authentication Specification*;

- Floor limit for magstripe card/chip card transaction;

- Application Identifier (AIDs);
- Application Version Number;
- Random selection parameter (for terminals with both online and offline capabilities);
- Processing restrictions parameter.

8.3.1 Implementation activities

Terminal management system modification includes the following operation and technical activities:

- Operation

  Terminal management system shall be able to verify terminal data elements after the deployment of terminals. In order to satisfy needs of local market, IC card modification and other chip applications, terminal management system must make necessary adjustment to the terminal data elements.

- Technology

  The acquirer shall extend the structure of terminal management system to ensure the support of chip data elements (for standard and other chip applications) and the flexibility of verification. If internal resources are not sufficient, the terminal management system can be modified together with the vendor.

8.4 Interface between Terminal and Acquirer

IC card introduces new data elements for the interface between the terminal and the acquirer. The terminal must be able to accept effective cards and transaction data, and to convert data according to the interface data format.

Information required by the interface between the terminal and the acquirer can be determined based on the local market situations. However, the interface between the acquirer and UnionPay System must be consistent with requirements in the *Technical Specifications on Bankcard Interoperability (Version 2.1)*.

The acquirer must upgrade messages of the interface between the terminal and the acquirer to support newly-added data elements. As the interface information is proprietary for each bank, UPI will not interfere with the processing. In order to facilitate the acquirer to understand data required by these messages, IC card specifications define the minimum set for the data of interface between the terminal and the acquirer.

It is recommended that the acquirer determine modification work required by the interface between the terminal and the acquirer based on reference and analysis of UnionPay System messages.

Note: If the acquirer can obtain field 35, then must submit this information, at the same time correctly submit each tag according to specification requirements on field 55, sending neither too much or too little data.

8.4.1 Fallback

Fallback refers to chip card executing non-chip transactions (magnetic strip transactions) on chip compliant terminals. The acquirer can recognize the following message formats as indicating fallback transactions:

- field 22 "POS entry format code (PAN entry format)" is not 05, 07, 91, 95, 96, 97, 98, indicating a non-chip card read format.
- field 60.2.2 "terminal read capability" is 5 or 6, indicating a terminal that can read chip cards.
- field 60.2.3 "IC card condition code" is 1 or 2, indicating exception while terminal is reading IC card.
- field 55 (IC card data field) is not present

When a chip card is accepted via the magnetic stripe at a chip-capable terminal, the transaction is referred to as a fallback transaction. These transactions are deemed less secure because magstripe acceptance circumvents the control and risk management protection available with chip acceptance.

The acquirer needs to specify fallback transactions in information of the interface between the terminal and the acquirer. As the information formats are proprietary, local market needs must be taken into account for determine how to change. Format of the interface between the terminal and the acquirer needs to include sufficient information so that the interface information of the acquirer and UnionPay System can correctly indicate fallback transactions.

8.5 Host System

Acquirer host system implementation of IC card migration can obtain more valuable returns. Acquirer host system needs to support some new data fields for transaction message and settlement files.

8.5.1 IC Card Migration Requirements

In UnionPay System, for online transaction messages, most new data is stored in “IC Card data field” (field 55) collectively; for settlement information, most new data is stored in *Technical Specifications on Bankcard Interoperability (Version 2.1).*

Whether card submits Segment 2 uses the following rule

- Card submits Segment 2 during electronic cash offline purchase transaction, but in electronic cash offline refund transactions the segment would not appear.

- When transaction medium is IC card, card submits Segment 2, if it is another medium then it is not submitted; activities required for acquirer host system to undergo IC card migration include: The technical team needs to evaluate the impact of full migration on the host system. The acquirer shall decide different technical support based on different circumstances such as in-house development, host system vendor, or third party processing service provider:
  - ✓ In-house development: Fulfillment of full migration is of great significance to the acquirer. There shall be a formal project plan to be initiated. At first, personnel must be designated from technical departments with more staff to incorporate a full around technical team and one project manager and this team shall be dedicated to the project. Similar to all formal projects, there must be complete establishment processes for project plan, regular meetings, meeting time and meeting themes list. Business requirements, technical requirements, design document, test and quality guarantee processes are mandatory. Development time term depends on concrete situations of the acquirer. Appendix A provides the template for project implementation plan.
  - ✓ Host system vendor: Most acquirers use the customized software provided by host system vendors to satisfy their specific needs. These acquirers need to install vendor software and make necessary adjustments to satisfy relevant requirements. It is suggested that the entire system must be tested thoroughly after software customization is completed.
  - ✓ Third party processing service provider: If the transaction is acquired by any third party processing service provider, the processing system needs to be tested.

After development is completed, the system must be tested to guarantee correct processing of transactions.

Whatever the special realization process is, all the acquirers that realize full migration must pass through host authentication of UnionPay System. For details, please refer to *Section Acquirer Host Authentication of this guide*.

8.5.1.1 EC Transaction

Standard EC transactions are all offline purchase transactions. The settlement information it forwards shall at a minimum include EC balance and EC issuer authorization code.

IC card terminal that supports EC function usually forward the data element - EC balance - when it performs online processing of standard IC card application request, so that the issuer will perform relevant processing such as automatic load check and risk monitoring. The data element is stored in the IDD part of the original tag '9F10', which will not cause additional impact on transmitting of messages.

8.5.1.2 qUICS Transaction

Given that standard IC card applications are supported, the system modification required by qUICS transaction processing is not much. The main work is to support new values of the existing data fields:

- Point of service entry mode (field 22)

  PAN entry mode is 07-qUICS entry.

- Terminal reading capability (field 60.2.2)

  6-The terminal has contactless card reading capability.

IC card data field

There may be additional new tags or data. For example: for qUICS online transactions, if the card supports to return offline available purchase balance, corresponding value will be obtained at the issuer discretionary data (IDD) part of the issuer application data (Tag '9F10'). The issuer can supervise and analyze offline fund usage situations based on this to enhance risk management.

8.5.2 Fallback

When a chip card is accepted via the magstripe at a chip-capable terminal, the resulting transaction is referred to as a fallback transaction. These transactions are deemed less secure because magstripe acceptance circumvents the control and risk management protection available with chip acceptance.

As the issuer may adopt different processing to Fallback transactions, the acquirer needs to indicate Fallback transactions in the messages sent to UnionPay System.

Activities related to Fallback include:

- Technology

  The system shall be modified so that the Fallback transactions can be identified in the acquirer's host system messages.

- Risk management

  The acquirer shall monitor the occurrence situations of Fallback to avoid plenty of Fallback transactions at an incomplete terminal and to prevent the merchant from violating operating regulations or fraud through collusion.

9 Public Key Management

This Section describes public key management activities when the acquirers' terminal supports offline data authentication. For UICS terminals that do not perform these functions, public key management activities are not mandatory.

Note: Prior to the loading of production key, the acquirer shall remove all testing keys to avoid offline data authentication failure.

Note: SDA, DDA and CDA all adopt the same RSA public key technologies.

For acquirers, it is not necessary to perform new DES key management activities. In addition, existing online PIN verification does not need any change for the moment.

9.1 Root CA Public Key

These public keys are used to support SDA, DDA and CDA. Any payment scheme has six effective private and public key pairs at most. Therefore, terminal management system stores at most 6 public keys for each RID. For acquiring root CA public key, please contact a UPI representative.

In order to upgrade to new key orderly and to promote revocation of old keys, all IC card terminals shall be able to support key updating. The data integrity must also be verified after configuration.

Table 11 UICS Root CA Public Key Distribution Process

Step	Actions
1	The acquirer fills "Member Root CA Public Key Certificate Application Form" and submits it in emails with encrypted signature.
2	UnionPay Root CA Authentication Management Section approves the application.
3	UPI auditor instructs the acquirer's security officers to download root CA public key certificate through emails with encrypted signature.
4	The acquirer's security officers verifies root CA public key certificate and informs UPI auditor of the verification results.
5	The acquirer will load valid UICS root CA public keys to terminals. During the transmission process, the public key information must be guaranteed data integrity and information source validity.

Note: For details of UICS root CA public key application, please refer to the Financial *IC card Debit/Credit Application Root CA Public Key Authentication Specification*.

9.2 Implementation activities

Policy, Operation and technical activities related to public key management include:

- Policy

UPI will regularly re-evaluate the expiry date of root CA public key. The evaluation result may be that revoking the current key, introducing new key (key length specified) and prolonging life cycle of the current key. Upon any change to the key, UPI will inform members of policy details, including key length, public key indicator, expiry date, planned revocation time schedule, etc.

- Operation

Locate root CA public key and public keys of other payment schemes and cooperate with the technical department to guarantee that these keys are loaded onto terminals correctly.

Note: Each payment scheme shall have its independent public keys.

According to EMV specification, the terminal supports at most 6 public keys for each payment scheme and the length of key can reach maximum 1984 bit. Now UPI supports four kinds of public keys with different lengths (1024/1152/1408/1984 bit). These keys must be stored in the terminal management system in correct format. The terminal management system shall confirm that it is a valid terminal which it communicates with before it downloads keys.

The acquirer must remove expired and revoked keys. For the expiration date of the current root CA public key, please see the latest periodic announcement on the UnionPay official website.

As these keys have life cycles , they shall be removed from terminals upon expiration. If one key is disclosed, it will be revoked in advance; the disclosed key must also be removed from the terminal. In addition, the new keys need to be loaded onto the terminal. The operation department must establish the processing flow for key upgrading and key replacement of terminals. UICS defines basic requirements on key replacement and revocation. For details, please refer to the *UnionPay Integrated Circuit Card Specification – Basic Specification Part 4: Debit/Credit Application Security Specification*.

The acquirer shall cooperate with the vendor to guarantee that terminals support all valid keys and key lengths. UICS, UPI, and CFCA make an evaluation of root CA public key security on an annual basis, deciding whether to extend the root CA public key certification. If root CA public key is extended, the acquirer must download the latest root CA public key to terminals on a timely basis.

- Technology
  - ✓ To obtain root CA public key certificate, verify it and extract root CA public key;
  - ✓ To load root CA public key onto the terminal before deployment;
  - ✓ To upgrade and replace current keys through downloading if necessary;
  - ✓ To guarantee data integrity of the downloaded key information;
  - ✓ To maintain configured key status through the terminal, for example: to ensure that the terminal has not lost key or redundant key and that all keys are valid;
  - ✓ The terminal vendor shall be able to load key to the terminal in advance during implementation.

10 Acquirer Host Authentication

This section describes the host authentication for acquirers that as required for implementing IC card migration.. Host authentication is mandatory for acquirers.

Note: "UPI Operating Regulations" requires: acquirers that accept UnionPay chip cards should realize complete migration.

This document does not involve other test items related to IC card migration, such as terminal, internal system, back-office process and front-office process, etc. As the scope IC card migration involved is broad, a thorough test shall be performed on various components. The work shall also be included in the acquirer’s test plan.

UPI will provide test tools to assist the acquirer to conduct host authentication. For further knowledge of UPI test tools, please contact a UPI representative.

10.1 Authentication Environment

Once internal tests on the modified system are completed, the acquirer needs to perform host authentication. The first step of the authentication process is to ensure that the components required are in place:

- UPI test tools;
- Authentication script;
- Personalized chip test card;
- UnionPay System connection;
- Manufacture complete terminal.

Note: Before production keys are loaded, the acquirer shall remove all testing keys to avoid offline data authentication failure.

Please contact the representative of UPI to obtain authentication script and other authentication materials.

UPI suggests that the acquirer test the connection with UPIS with magstripe card test transactions one to two weeks before online testing, so that any connection problem can be detected and solved in time.

10.2 Authentication Process

This section summaries the host authentication process on the acquirers. The acquirers must perform a serial of transactions according to the authentication test script to prove that the acquirer’s host system is able to send and receive new data fields of each message.

For acquirer network testing process please see Appendix C.

Note: UICS Debit/credit acquirer network access test also includes “terminal authentication”. If the acquirer adds a new IC card terminal model, or root CA public key changes, or terminal hardware/software changes, which may affect online connection, the system shall go through terminal authentication. For details about terminal authentication, please consult the representative of UPI.

11 Acquirer Back-office System Modification

This Section describes modification of back-office functions that support IC card migration plan. The modification tasks cover the following aspects:

- Interchange rate;
- Bookkeeping;
- Dispute resolution;
- Reporting;
- Internal staff training;
- Implementation activities.

11.1 Interchange Rate

Please contact the representative of UPI to obtain detailed information of service fee and network service expense.

11.2 Bookkeeping

IC card migration will not bring new fee types to the acquirer. The bookkeeping and fee types of UPI will also not change. For further details, please contact the representative of UPI.

11.3 Dispute Resolution

New IC card transaction data and processes will impact customer dispute resolution, chargeback, re-presentment and arbitration. Potential influence of such changes on dispute resolution needs to be analyzed. Dispute resolution allows the acquirer to respond to chargeback to the issuer and initiate re-presentment through providing transaction receipts. The acquirer also needs to analyze the impact of modification to relevant dispute resolution on the host and customer service system.

- Chip data can be used to respond chargeback transaction; therefore the acquirer needs to modify the system to support longer data storage, for example 180 days. In addition, information related to disputes can also be obtained through accessing chip data.
- Display method required by chip data shall be evaluated, for example in the form of statement or in the form of display screen.
- Confirm chip data that the system need to obtain, process, store or back up which have an impact on dispute resolution and modify system as required.
- Support new error code for IC cards in dispute resolution..

Table 12 New Error Code

Application Error Code	Description
6308	A problematic transaction, please refer to transaction certificate (TC) and relevant calculation data.
4558	Transaction certificate (TC) authentication fails.
4559	TC and relevant calculation data cannot be provided.

11.4 Internal Staff Training

The acquirer shall conduct trainings on chip card transaction processing to the merchant service related departments. Operation process training shall be conducted at the point of transaction with deployment of UICS terminals. Workload of training depends on how many departments are related to merchant service. The departments that interact directly with the customer shall be trained more extensively.

An all-around training plan helps to guarantee that the project can be launched as soon as possible. Training plans consists of the following tasks:

- To setup training objectives;
- To determine training requirements;
- To design training courses;
- To prepare training materials, operation guidelines and help manual;
- To adjust training requirements based on training progress;
- To establish training schedule;
- Personnel that needs training include:
    - ✓ Personnel at the headquarters;
    - ✓ Personnel at branches;
    - ✓ Personnel of customer services;
    - ✓ Operation personnel;
    - ✓ Bookkeeping personnel;
    - ✓ System development personnel;
    - ✓ Personnel of relevant branch affiliates.

After the initial trainings are completed, the staff must have the knowledge of how to solve problems. For support and suggestions on trainings, please contact a representative of UPI.

11.5 Implementation activities

This section describes policy, operation and technical activities related to interchange, bookkeeping, dispute resolution, reporting and trainings:

- Policy
  - ✓ To analyze product fee structure and decide whether fee standards shall be changed ;
  - ✓ To evaluate report requirements on new chip transaction data.
- Operation
  - ✓ To determine appropriate interchange rate and to evaluate financial impact on the merchant;
  - ✓ To inform the merchant of new interchange rate and fee structure if necessary;
  - ✓ To evaluate the impact of IC cards on dispute resolution rules:
  - ✓ To design and establish procedures to adapt to chip related changes;
  - ✓ To inform the merchant of any changes of the processing procedures;
  - ✓ To determine the modification requirements of customer service and dispute resolution system.
  - ✓ To decide which existing reports need to be changed and which new reports shall be added;
  - ✓ To establish and execute training plans covering all participants.
- Technology
  - ✓ To modify the system to support new interchange rate and fee structure (if applicable);
  - ✓ To enhance the existing reporting function and to design new IC card reports;
  - ✓ To modify customer service and dispute resolution system.

12 Merchant Support

This Section describes relevant tasks that shall be fulfilled to help the merchant support chip cards. Scope of merchant support associated with IC cards includes:

- Merchant agreement;
- Merchant services;
- Merchant system modification;
- Merchant training

12.1 Merchant Agreement

The current merchant agreement must be updated to adapt to the IC card migration. It is crucial to evaluate the impact of chip processing on merchant relationship. The following contents shall be changed to the merchant agreement:

- Terminal cost and installation, and changes to the fee structure;
- Additional authorization and settlement information;
- New reports;
- Expense and competition factor;
- Expectations of the merchant on accepting chip card, including evaluation on chargeback responsibilities;
- Changes to the card acceptance process.

Adjustments in the merchant agreement that are related to commercial requirements shall be issued in formal notification. In addition, the legal consultant shall review the agreement.

12.2 Merchant Services

During deployment of chip-capable terminals, more merchant services and support are needed. In order to maximize acceptance of IC debit/credit card products, the merchant service scope expansion must be carefully planned out.

12.2.1 Support on Merchant Implementation

When acquirers decide to deploy chip-capable terminals, it is essential to plan how to support the merchant to migrate to IC cards. The number of problems experienced by the merchant during migration will directly influence the success of the acquirer's plan. Therefore, full range preparation and strong support and training to the merchant are necessary. The following items need to be completed:

- To decide how to provide hardware and software installation support;
- To estimate possible problems and ranges and to prepare solutions in advance;

- To guarantee that the merchant's service personnel are well trained and can cope with requests and problems at the merchant's side;
- To arrange trainings for trainers;
- To consider hotline installation and provide customer consulting;
- To evaluate potential terminal modification contents under the merchant's environment, including appearance change, cashier desk change, electric upgrading, PIN keyboard placement and wiring changes;
- To determine the effect of the merchant's network interface on supporting chip data processing;
- To determine the terminal maintenance type, such as internal support or third party support.

12.2.2 Terminal Installation

The last step of terminal deployment is to provide operable terminals based on the merchant's situation. The terminal deployment plan needs to take the following aspects into account:

- To assist the merchant in setting the merchant name and location (tag '9F4E') correctly;
  - ✓ The Terminal shall fill in and submit merchant name and location (tag '9F4E') correctly according to transaction log format (tag '9F4F').
- To test all the terminal components to guarantee normal operation;
  - ✓ To perform basic functionality test on various components of the terminal;
  - ✓ To perform an overall test on the terminal and guarantee interoperability between components.
- To guarantee that all terminal data has been completely downloaded before transmission to the merchant, including application identifier (AID), application version number, terminal action code (TAC) and public key, etc. and that the terminals work normally.

Note: Before production keys are loaded, the acquirer shall remove all testing keys to avoid offline data authentication failure.

- To determine feasible deployment methods;
- To determine the deployment priority based on region or merchant creditability, etc.;
- To determine the delivery of supplies , e.g. printing paper, ribbons and equipment panels, etc.;

- To determine the required accessories, e.g. terminal base and power lines;
- To develop service agreement, service desk support and training;
- To provide terminal operation manual, to the merchant;
- To decide whether cardholder operation manuals are provided at the point of transaction. If provided, the contents such as prompt whether the cardholder is requested to enter PIN shall be included in the manual;
- Through onsite installation and testing, it must guarantee that root CA public key and other parameters related to chip have been correctly downloaded.

12.2.3 Ongoing Merchant Services

The merchant's service and support personnel must be prepared to reply inquiries from customers that are introduced to chip-capable terminal. In order to ensure effectiveness of support, it is recommended that the following measures can be taken:

- To determine available resources and types of inquiries;
- To determine the expected level of terminal support.

The acquirers shall provide suggestions to merchants by the hotline. The hotline phone number can be provided through the following ways:

- To post on the terminal;
- To list it in the merchant's settlement account reconciliation report;
- To include it in the merchant training materials or certain information materials;
- To include it in the brochure or address book prepared for the merchant.

12.3 Merchant System Modification

The project plan shall consider the impact of chip data application on the merchants. Modification of the merchant system shall be evaluated in terms of the following content:

- Interface between the terminal and the merchant host;
- Interface between the terminal and retail workstation;
- Internal terminal controller;
- Interface between the merchant and the acquirer's host;
- Merchant back-office system;
- Plan merchant network scale to satisfy the needs of transaction processing, obtaining, recording and backup.

Once modification is completed, sufficient time shall be reserved to test the final merchant system configuration.

12.4 Merchant Training

New functions are introduced to the point of transaction through deployment of IC card compatible terminals. The training plan shall focus on changes to the merchant operation procedures.

12.4.1 Training Topics

The training plan shall cover the following topics:

- General chip acceptance procedures;
- Cardholder application selection;
- Cardholder ID verification;
- Offline transaction and online transaction;
- Fallback transaction;
- Other transactions;
- Other application support;
- Terminal maintenance.

Some of the features above are specific to the chip card and the terminal. They are transparent to the merchant and the cardholder.

For detailed information of the functions above, please refer to *Chapter 6 and Chapter 7.*

12.4.1.1 General Chip Acceptance Procedures

The merchant must be trained on the differences between procedures for magstripe card transaction and chip card transaction:

- After the chip card is inserted into the chip reader, it shall remain inserted during the transaction. It is different from the way of magstripe card transaction where the merchant needs to swipe the magstripe card in a single motion..
- During the transaction process, the chip card must be kept in the terminal until the transaction is completed. If the chip card is removed early, the transaction will be terminated. It must be guaranteed that the terminal is able to display relevant prompt information when the transaction is completed, so that the merchant or client is able to take the chip card out of the reader upon noticing the prompt information.

- At the points of transaction where the cardholder has to insert the chip card by himself, the merchant shall explain chip card processing procedures to the cardholder. In addition, at unattended terminals (such as ATM or terminals activated by the cardholder), prompt information must be displayed to instruct the cardholder to complete each transaction step.

Due to the above changes, the acquirer shall evaluate whether it is necessary to prepare cardholder operation manual for merchants so that they can help the cardholder to complete chip card transaction correctly.

12.4.1.2 Cardholder Application Selection

The chip card usually supports multiple applications. If the terminal supports many applications, application selection shall be provided to the cardholder. Training for the merchant shall include descriptions on how the merchant explains application selection procedures to the cardholder. In addition, the merchant shall also be trained on how to instruct the cardholder to click appropriate buttons to selection desired application or account.

The cardholder application selection process usually depends on the definitions made by the issuer in the chip. The card and the terminal can automatically select the application with highest priority for the transaction and can also perform cardholder selection for the application supported by both parties, or confirm with the cardholder for each application based on priority.

The merchant also needs to note that not all transactions require application selection. Normally, only when the card and the terminal both support several applications, selection is needed. For example, when the card and the terminal both supports UICS debit card and credit card applications simultaneously, the cardholder is required to select the application. Therefore, the merchant needs to understand that certain transactions perform the application selection while some other transactions don't perform the application selection. This only represents different transaction types, instead of mistakes.

12.4.1.3 Cardholder Verification

The merchant and the cardholder shall understand that in an attended environment, the cardholder can confirm a transaction through signature or PIN entry mode; in an unattended environment, regardless of whether PIN entry is required, signature is meaningless.

Under the chip acceptance environment, the cardholder verification method (CVM) required to complete transactions is decided by the terminal and the card. The selection of CVM is beyond control of the merchant and the cardholder.

Interactive decision process between the terminal and the card and the final selection are decided based on some data elements of specific transactions, e.g.:

amount, domestic or international transactions, online or offline transaction and other transaction parameters.

CVMs discussed below include:

- Signature;
- PIN;
- CVM not required.

12.4.1.3.1 Signature

According to the *UPI Operating Regulations*, UPI cards shall have both magstripe and signature bar.

Signature is still a default method for international ID verification for cardholders. It is also the default method of card transaction confirmation in many regions. Signature verification under chip card environment and that under Magstripe card environment are consistent.

12.4.1.3.2 PIN

PIN entry for cardholder ID verification is convenient and safe. This makes PIN entry popular with IC card transactions. For places that deployed with PIN entry keyboards, training shall cover the following points:

- Interaction between the card and the terminal will decide appropriate cardholder verification method (CVM) and whether PIN entry is prompted. As it is the card that decides whether PIN entry is required during each transaction, it is not a mistake if the terminal does not prompt PIN entry. If the chip card requires PIN entry, the terminal will make prompt. The merchant shall not require the cardholder to enter PIN unless the terminal makes such prompts.
- When the cardholder is required to enter PIN, PIN entry process must be kept confidential.
- If a transaction is performed through PIN entry, it is suggested that the signature bar should not be printed on the receipt. Refer to *Section 6.4 Cardholder Verification* for more information. The merchant shall note that the cardholder shall not be required to sign if the receipt has no signature bar.

Note: In some regions, offline PIN verification and cardholder’s signature maybe required for transactions with an amount exceeding certain limit. Please contact representative of UPI for local market requirements.

12.4.1.3.3 CVM Not Required

The issuer is able to complete transactions through other verification mechanism even if it does not require the cardholder to sign or to enter PIN. When the terminal and the card both allow “CVM not required” as the cardholder verification option, “CVM not required” will be a valid CVM option.

This method is usually applied for unattended terminals. The issuer can select this method to complete quick offline transactions. Anyhow, even if the card instructs “CVM not required” for certain terminal types, the terminal can still select the CVM in magstripe card transaction as its default CVM (such as to use signature on POS, or to use online PIN on ATM) and provide security protection to transactions.

This method can be taken as preferred CVM for low value payment.

12.4.1.4 Offline & Online Transaction

At the merchants where online /offline terminals are deployed, the cashiers shall be trained so that they can understand certain transactions are processed offline while the others are accepted for online processing. They do not need to focus on differences between online and offline processing or regard such differences as errors. However, they shall at least understand that offline transaction is faster than online transaction.

12.4.1.5 Fallback Transaction

A Fallback transaction refers to a transaction that when the chip card or the chip-capable terminal cannot operate normally, the terminal instructs the cardholder to swipe the magstripe on the chip card.

UICS terminals shall support normal chip card transactions and Fallback transactions. The terminal shall first try to process the transaction by chip. Only when the chip or the chip reader cannot work normally, a Fallback transaction can be performed. The terminal device is not allowed to prompt to use the magstripe by skipping the chip authentication control.

When the chip or the chip reader cannot work normally, the terminal will obtain magstripe information of the chip card through the magstripe reader. In this case, the terminal will read the service code from the magstripe information and prompt the merchant to perform operations as a chip card. The acquirer needs to arrange Fallback training for the merchant. The acquirer must abide by certain procedures. Typically, the cashier is offered many chances to read the chips. Before the magstripe is applied for the Fallback transaction, the cashier will be prompted to swipe the card.

The merchant must understand that for declined chip card transactions Fallback transactions should not be attempted. Declined chip card transactions cannot be completed through magstripe or any other methods. In this case, the transaction will be declined by the issuer.

When a chip card transaction is declined or fails, the merchant can consult the cardholder whether other payment types can be applied.

12.4.1.6 Other Transactions

Processing procedures of transactions of refund, reversal and cancellation are the same as the current ones. But they are performed through chips rather than magstripe. If necessary, other card security characteristics must be checked at the point of transaction.

12.4.1.7 Other Application Support

If the acquirer supports other chip applications, such as loyalty program or stored value scheme, the merchant shall also be trained on these new operation procedures. The acquirer needs to include these activities in the training plans.

12.4.1.8 Terminal Maintenance

Trainings shall include contents related to daily terminal maintenance, magstripe and chip reader cleaning and interference item clearing, etc. In addition, it is suggested that the terminal shall be maintained and checked regularly to guarantee normal operation and proper location of various components and to prevent accidental damage and data loss.

12.4.2 Merchant Training Plan

The merchant training plan usually includes the following content:

- To establish training objectives;
- To determine training requirements;
- To design training courses;
- To prepare training materials;
- To train the trainers.

12.4.3 Training Materials

During the training process, training supporting materials need to be provided. Usually the following materials are provided for merchant training:

- Training introduction;
- Operation guidelines;
- Quick reference guidelines;
- FAQ for merchants and cardholders;
- Contact information of the Merchant Service Department.

Customer training materials help the merchant solve common problems from cardholders. Through initial trainings, the merchant’s personnel shall be able to seek appropriate methods to solve problems.

If many Fallback transactions occur or certain operation problems repeatedly occur, it is necessary to arrange further training.

The merchant’s training materials and training requirements shall be evaluated so that the merchant can establish its own training mechanism.

Appendix A (Informative Appendix) Project Implementation Recommendations

Implementation activities and time required for realizing IC cards depend on concrete requirements of the acquirers. Generally, the time required to complete IC card migration plan is 9 to 18 months. At the preparation stage, full understanding of features and benefits of IC card products and how to promote commercial needs of the acquirers is significant for implementation of the entire project.

IC card migration plan involves a broad range. It will influence the bank's clerks, merchants, terminal vendors, business procedures and processing system. It has strict requirements on project management. A cross function team needs to be incorporated which should consists of the capability to deal with several tasks simultaneously.

The appendix is used to help the acquirer to plan the IC card migration project and to establish a detailed working plan, covering:

- Key success factor;
- Project organization;
- Implementation plan;
- Summary list of project tasks.

A.1 Key Success Factors

During the planning phase, project objectives, scope and success standards shall be determined. Project leader, organizer, organization structure and departments involved need to be determined. Successful completion of this work will lay the foundation for smooth implementation of the entire project.

Clear and specific objectives can lead the project to the correct directions and make participants to focus on major problems. For example, if one of the major objectives of the acquirer is to provide bonus application based on chip to the merchant, then it shall be guaranteed that the terminal selected and the software can support the function, while other irrelevant optional functions do not need much attention.

The acquirer can decide the method to realize the plan based on the project scope. The acquirer shall determine the appropriate realization method based on device, capital and market scale. It will be much easier to try 500 cards at two self-service restaurants than try 2500 terminals or 50,000 cards.

During the project startup phase, the success standards shall be defined and approved by all participants and the main leaders. At each project stage, members can participate in difficult problem solving and determination of communication requirements based on these standards. Success standards are also the basis of quality guarantee of implementation and user acceptance. During product launching and closing preparation, the standards will be applied to appraise success or not. Finally, the standards will be applied to make conclusions for the project and to measure whether the project is completely successful.

Whatever the scale of project organization is, several key success factors shall be taken into account:

- Leader and executor

  There must be an excellent leader who shall be capable to manage different teams and to fully promote spirits of various parties to guarantee project success. Person in charge with strong execution capabilities is also very important to project organization.

- Roles and responsibilities

  The entire project organization involves several domains. Therefore, before implementation process is started, role and responsibility of each domain shall be defined. Many activities have cross-independence or successive relationship. Therefore, each team shall be clear about its own role to the entire project.

- Preparation work

  In order to successfully implement IC card migration plan, sufficient preparation is critical. Development of many activities greatly depends on determination of business policy.

- Decision and management committee

  As this project involves many business departments within the bank, there can be problems for which agreement can not be reached during business requirements discussion. Therefore, a special Decision and Management Committee is helpful to collectively coordinate and solve difficult problems and to avoid impact on project progress.

- UPI suggests that the Committee shall remain over the entire life cycle of the project. The Committee shall include principals of various parties. They do not have to participate in project everyday, but to provide instructions to project implementation, arbitrate policy that can not be reached by the project team, and provide capital and resources required by project, etc.

A.2 Project Organization

Most member banks build a dedicated working team to manage all aspects from planning to product launching. The team consists of representatives of various bank departments that are impacted by IC card migration. Each team member provides professional suggestions to relevant fields and assumes relevant responsibilities.

The experts required for the project shall participate in the project as early as possible. Of course, this does not mean that all members need to be dedicated to the project. The part below describes the functions, roles and responsibilities of a typical IC card migration team.

A.3 Project Manager

The Project Manager is responsible for overall operation, milestone and timeline of the project, and daily management, intra-team coordination problem, project activity and task tracing, project document maintenance and distribution, meeting scheduling, and reporting and coordination to the Management Committee. The Project Manager shall also assume the role of main contact to UPI. In order to guarantee that the Project Manager can concentrate energy to solve major problems, assistants can be designated to assume part of the Project Manager's responsibilities.

A.4 Project Team

Project team members will report to the Project Manager. The Project Team usually consists of members of the following aspects:

- Card acceptance product manager

  Card acceptance product manager will supervise the structure of the migration plan from the aspect of business and decide the acceptance functions and the business policy that should be realized. The card acceptance product manager is also responsible for contact and management of vendors, and may also be responsible for the management of newly-added keys.

- Marketing

  The marketing personnel analyze the market prospects for cardholders and merchants based on potential product services, submit relevant reports and establish marketing policy.

- Legal

  The IC card migration plan usually requires the acquirers to change the cooperation agreement with merchants. In addition, they also involve new contract signing with vendors, or amendment to original contracts. The work is assumed by the legal department.

- Security and risk management

  The security and risk management personnel are obliged to provide professional knowledge of risk control and information security, management of symmetric key and asymmetric key, to monitor correctness and security of key generation, and to guarantee safe transmission of keys.

- System development

The system development personnel are obliged to implement system modification related to IC card migration.

- System and network management

  The system and network management personnel are responsible for test environment, installation, modification and maintenance of the production environment. Works on this regard shall be arranged for implementation on time to avoid any impact on the project progress.

- Business management

  The business management personnel are obliged to establish IC card related business process, such as dispute resolution and customer service.

- Merchant marketing and service

  The merchant marketing and service personnel are obliged to contact the merchant and to promote VIP merchants to participate in the plan. During the entire implementation process, the personnel shall coordinate and assist the merchant.

- Training

  The training personnel are obliged to establish training plans, to prepare training materials, and to provide focused trainings to employees of different positions.

- Documentation

  The document management personnel are obliged to prepare and amend operation and technical documents.

- Quality management

  The quality management personnel shall execute all necessary tests and guarantee correct system running after modification.

- User acceptance

  The user acceptance personnel shall perform tests and guarantee that the transformed system can satisfy business needs of the end user.

A.5 Implementation Plan

The team will establish the implementation plan based on the project objectives. The plan shall cover activities occurred during the entire project life cycle. Although the concrete format and contents may vary based on different bank plans, the following contents shall be included:

- Major milestones expected to achieve;

- Major problems to be solved;
- Sequence and time schedule of key events;
- Proper descriptions on implementation tasks.

After the implementation plan is completed, the team evaluates each major function module, prepare an entire detailed set of tasks and assign these tasks to each team member based on the work plan, and specify relevant responsibilities and time requirements.

The following "Summary List of Project Tasks" can help the acquirer to establish UICS migration implementation plan.

A.6 Summary List of Project Tasks

Step	Task
1	Project preparation activities
1.1	To obtain approval from major leaders and establish a management team
1.2	To determine authority scope of the management team
1.3	To appoint the management team and start operation
1.4	To discuss with representative of UPI about impact of IC card migration
1.5	To train members of the team on IC card related knowledge
2	Decision
2.1	To define commercial opportunities
2.1.1	To describe current card product business mode – audience, process and techniques
2.1.2	To determine business module and organization value of IC card application
2.1.3	To judge whether the technical conditions are mature or not (internal & external)
2.1.4	To define business objectives

Step	Task
2.1.5	To describe possible benefits
2.1.6	To specify business requirements
2.1.7	To specify commercial opportunities and risks
2.1.8	To describe market environment and external impact
2.1.9	To describe IC card business requirements
2.2	To seek approval
2.2.1	To report IC card migration plan to major leaders
2.2.2	To approve suggestions of the management team
2.2.3	To apply for budget and human resources
2.3	To incorporate the project preparation team
2.3.1	To designate competent business and technical personnel to the project team; if necessary, third party manufacturer, outsourcing manufacturer, UPI or external consultants can join the project team.
2.3.2	To determine the authority scope of the project preparation team
2.3.3	To obtain IC card related files
2.3.4	To train members of the team
3	Preparation (planning and startup)
3.1	To analyze business solutions
3.1.1	To describe options and recommendations for business solutions – audience, process and techniques

Step	Task
3.1.2	To define IC card security policy
3.1.3	To describe the implementation method – self-realized, customized, outsourced or composite
3.1.4	To analyze impact of IC card migration on original businesses
3.1.5	To describe possible benefits – direct financial benefits, business process, customer's perception and organization innovation
3.1.6	To obtain the template of IC card migration implementation plan from UPI
3.2	To establish the project management plan
3.2.1	To decide the implementation budget – personnel, techniques, third party manufacturer, device, tool and other expenses
3.2.2	To define key success factors
3.2.3	To establish a quantified business solution
3.2.4	To describe how to manage, evaluate and statistic new benefits
3.2.5	To define internal and external communications and training policy
3.2.6	To define checking and testing policy
3.2.7	To perform risk analysis
3.3	Project planning
3.3.1	To evaluate the business mode and business scheme
3.3.2	To determine project objectives and scope
3.3.3	To determine work flow

Step	Task
3.3.4	To determine resources required (include how to obtain resources)
3.3.5	To describe project management methods
3.3.6	To evaluate change and level of the expected difficulties
3.3.7	To perform project risk evaluation and to establish risk management plan
3.3.8	To establish communication and training plans
3.3.9	To establish check and test plans
3.3.10	To form the project management plan based on the above contents
3.4	To start the project
3.4.1	To assign work flow resources
3.4.1.1	To formally incorporate project team
3.4.1.2	To train project team members
3.4.1.3	To execute project management methods
3.4.1.4	To execute plan control process
3.4.1.5	To establish the communications plan
3.4.1.6	To establish unit test, integrated test, certification test, user acceptance test, etc.
3.4.1.7	To establish project-responsible management measures
3.4.1.8	To execute measures for risk and emergency management
3.4.2	To establish benefit model

Step	Task
3.4.3	To execute benefit management method
3.4.4	To start various work flows
4	Full migration of the acquirer
4.1	To select vendors
4.1.1	To establish and release RFI/RFP
4.1.2	To evaluate returned information
4.1.3	To select vendors
4.1.4	To negotiate and decide the tenders
4.1.5	To manage vendors
4.1.6	To receive products delivered by vendors
4.2	To select terminals
4.2.1	To decide functions and security requirements of terminals
4.2.2	To analyze whether the terminal's specifications can satisfy various requirements
4.2.3	Terminal vendors provides quality management services
4.2.4	To confirm that terminal vendors have passed the certification of Bank Card Test Center
4.3	To modify the acquirer system
4.3.1	To modify the transaction processing system
4.3.1.1	To modify the terminal management system

Step	Task
4.3.1.2	To modify interface between the terminal and the acquirer
4.3.1.3	To modify the host system
4.3.2	To modify the business processing system
4.4	To obtain root CA public key
4.4.1	To apply for root CA public key from UPI
4.4.2	To safely transmit and verify root CA public key
4.4.3	To load root CA public key (production/testing) to terminals
4.5	To perform system network access tests – terminal tests
4.5.1	To perform terminal offline tests (TFT)
4.5.1.1	To apply UPI test card for tests
4.5.1.2	To submit test records
4.5.1.3	UPI reviews the test results
4.5.2	Terminal integrated test (TIT)
4.5.2.1	To use UPI test cards and simulators for tests
4.5.2.2	To submit test logs
4.5.2.3	UPI reviews the logs
4.6	To perform system network access tests - host tests
4.6.1	Offline test part

Step	Task
4.6.1.1	To use UPI test cards and simulators for tests
4.6.1.2	To submit test logs
4.6.1.3	UPI reviews the logs
4.6.2	Online test part
4.6.2.1	To perform online test certification
4.6.2.2	UPI issues test reports
5	Internal test
5.1	To determine test policy
5.2	To establish a detailed test plan
5.3	Unit test/integrated test/user acceptance test
5.3.1	To establish acceptance standards
5.3.2	To prepare test tools
5.3.3	To prepare test cases
5.3.4	To prepare test data
5.3.5	To construct the test environment
5.3.6	To execute the test plan
5.3.7	To check and verify test results
5.3.8	To perform regression test

Step	Task
5.3.9	To pass the acceptance testing
6	Merchant support
6.1	To determine current merchants participating in the migration and develop new merchants
6.2	To support merchant implementation
6.3	To modify MIS merchant system (if necessary)
6.4	To train merchants
7	Product launching
7.1	To determine the product launching policy
7.2	To establish the product launching plan
7.3	To establish the device delivery plan
7.4	To prepare the technical environment
7.4.1	Network connection
7.4.2	Environment test
7.4.3	System installation
7.5	To prepare for production launching
7.5.1	To plan and implement the production launching rehearsal
7.5.2	To import production launching data
7.5.3	Operation training

Step	Task
7.5.4	To test the disaster backup system
7.6	Production launching
7.6.1	To establish production launching schemes and expected results
7.6.2	To prepare the verification testing data
7.6.3	To execute production launching plan
7.6.4	To verify the correctness of IC card application
7.6.5	To pass the acceptance testing of the project by business departments
7.6.6	To transfer the project to the operations maintenance department

Appendix B (Informative Appendix) Basic Principles for Distribution of Root CA Public Key to Terminals

B.1 Basic Principles

In order to guarantee that the acquirers can safely transmit the verified root CA public key information to terminals, the acquirer must comply with the following rules:

- When the root CA public key is transmitted by the acquirer to the terminal, the terminal device must verify data integrity and authenticate the information source of the transmitted public key information. For example, digital signature or MAC authentication code can be applied during the transmission process.
- The acquirer can select authorized organizations to manage part of the terminals. The authorized organization can be an organization that has a business agreement with the acquirer and manages part of the terminals or data receiving devices of terminals, or an organization that has a business agreement authority with the acquirer and is responsible for maintenance of part of the terminals or data receiving devices of terminals.
- When the root CA public key is transmitted to the authorized organization by the acquirer, data integrity of the public key information transmitted must be verified, while the information source must be authenticated. When the root CA public key is transmitted from the authorized organization to the terminal, the same process shall also be conducted. A terminal data receiving device must be subject to effective access control when it receives root CA public key data or during the manual control process.
- When the verified root CA public key is transmitted to the terminal, data integrity must be guaranteed. This rule guarantees that the verified root CA public key data received by the terminal is complete. For example, Hash value of the public key information shall be verified; or effective inspection or authentication shall be implemented.
- It shall guarantee that the newly-introduced root CA public key information is loaded onto each terminal before the public key effective date. It must guarantee that the newly introduced root CA public keys are loaded onto each terminal before the circulating date of financial IC card corresponding to the root CA public keys, so that the financial IC card corresponding to the root CA public key can be applied when it becomes effective. The acquirer must be able to audit this requirement.
- It shall guarantee that expired or revoked root CA public keys are removed from each terminal or stopped from application within six months after such expiration or revocation. The terminal shall be able to identify expired or

revoked root CA public key. The acquirer must be able to audit this requirement.

- The acquirer must be able to decide which IC card payment system root CA public key are in use for each terminal within a reasonable time frame.
- The acquirer must be able to report the current key length supported by each terminal. The current supported key length must be synchronized with all the root CA public key length released by UPI within the effective term, so that all IC cards within UPI financial IC card public key system can be supported. In addition, the terminal must effectively prevent fraud with overdue root CA public keys.

B.2 Other Suggestions

- In order to optimize operation efficiency, when one key needs to be downloaded or removed, the terminal management system shall be able to inform all affected terminals automatically.
- In order to reduce expenses of manual key replacement, automatic notification of key replacement should be implemented. This notice can be realized through authorization response, batch settlement conformation or end of day response, etc., or be initiated by the terminal management system. In addition, the terminal can take the initiative to ask the terminal management system for a clear update.
- Terminals and terminal management system shall arrange automatic updating for impacted keys. It is difficult to perform manual management on large numbers of terminal keys. Therefore, it is suggested that once the notice is received; the system automatically performs a preparatory processing procedure and complete key updating in a timely fashion.
- The issuer shall comply with IAC settings recommended by UPI. If the card asks for a key but the terminal does not have it, there will be a SDA/DDA failure: the transaction will take further actions based on IAC and the issuer will decide how to proceed with the transaction. In this case, optimal settings of IAC shall be to instruct the transaction to request online processing.

Appendix C (Regular Appendix) Procedures of Network Access for UPI IC Card Acceptance

To initiate the functions of IC Card, the Acquirer needs to follow the procedures similar to those of Magnetic Stripe Card. That is to say, the network participant needs to pass the function tests first (offline test and online test), then apply for network participation.

C.1 Testing Procedures for Network Participation

The testing procedures for IC Card network participation are as follows:

Off-line Test Phase:

The network participant files an application for an offline simulation tool, and the CUP issues the authorization code and dongle or any other simulation tools. The network participant uses the offline simulation tool to conduct offline testing and generate an offline testing report and transaction log following the tests.

On-line Test Application and Acceptance Phase

The materials required for on-line test application include:

- On-line test application and authority list.
- Offline-testing report and transaction log

An institution at head office level transmits the above materials to banktest@unionpay.com for online test application. An institution under a branch transmits the above materials to the local branch of CUP for online test application. The information center arranges an online test within 4 work days upon receiving and approving the online test application.

On-line Test Phase:

Personnel from the information center assist the institution at head office level in completing all items required for online test and issue an online testing report, which, upon being examined and approved, will be delivered to the network participant and the business management department.

Personnel from the branch of CUP assist the institution under the branch in completing all items required for online test and issue an online testing report, which, upon being examined and approved, will be delivered to the network participant and business management department.

Network Participation and Operation Phase:

Following the tests, an operator from the information center discusses the network participation and operation time with the network participant and completes such work at mutually agreed time.

C.2 Special Instructions

For CUP IC Card network participation, the participant shall be clear about the following:

- The offline simulation tool application, online test application and authority list may be downloaded at the CUP official website.
- Requirements for test card: A test card from the bank card testing center shall be prepared, as required in the online test application, for online transaction test.
- The offline tests of internet participants do not include the E-Cash offline consumption, settlement document and error handling tests; the on-line tests include tests upon all transactions opened, including E-Cash offline consumption and return, settlement document and error handling tests.
- In network participation tests, the offline test cases shall be basically consistent with online testing cases.